

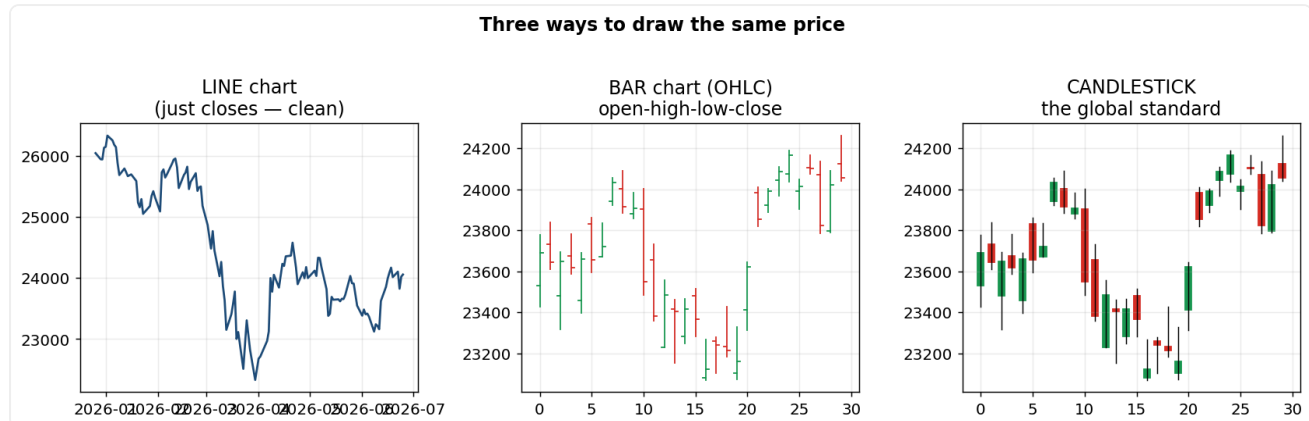
Technical Analysis Illustrated Handbook

Every topic: the chart, why it works, and the market psychology behind it.
Built to answer the interviewer's follow-up question.

Prepared for Saikumar Kaleru — Technical Research Analyst prep

PART 1 — Foundations

Three ways to draw price



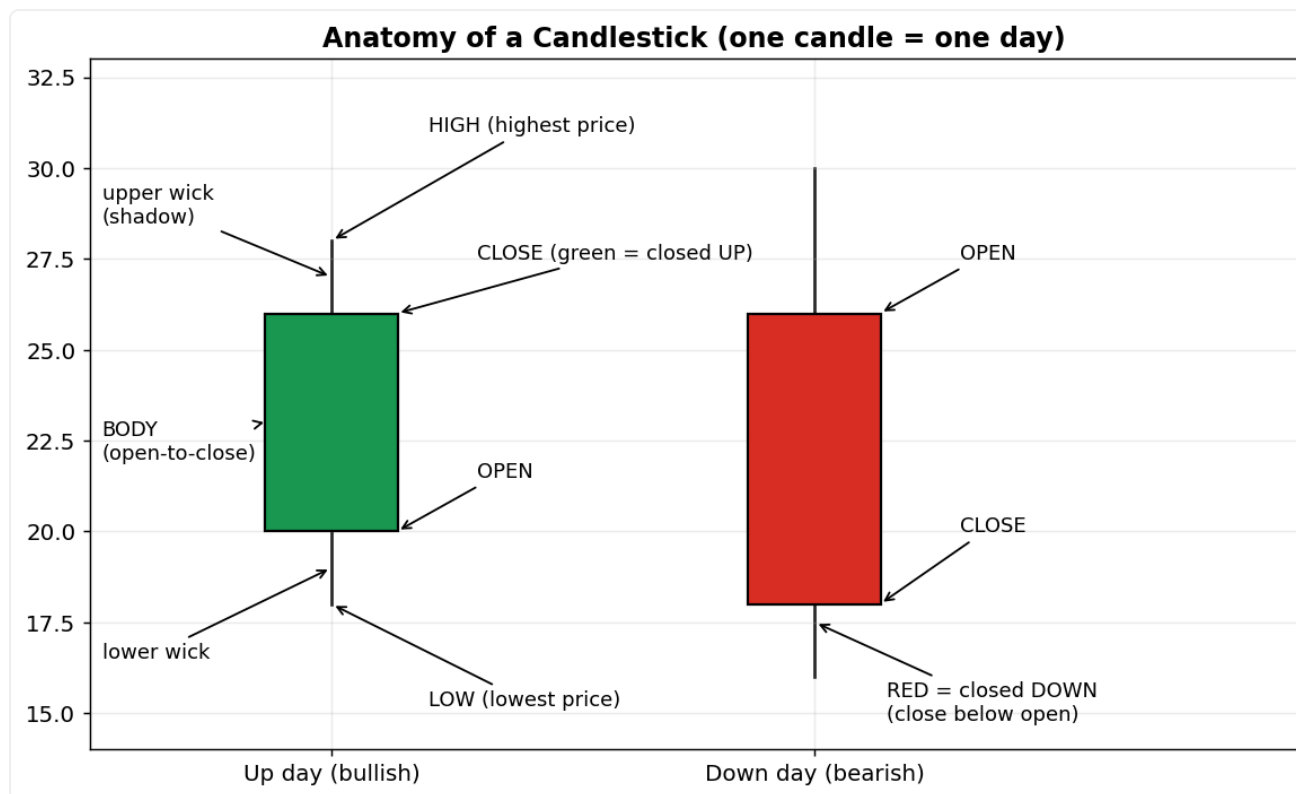
What it is: The same price history shown as a line (closes only), a bar (OHLC), or a candlestick chart.

Why it works: Candlesticks encode four prices (open, high, low, close) plus the winner of the day in one visual symbol, so you read the balance of power at a glance — that's why they became the global standard.

The psychology: Each candle is a snapshot of a battle between buyers and sellers. A big body = one side dominated; small body or long wicks = conflict and indecision. Charts work because they are a picture of collective human emotion, and emotion repeats.

How to use / trade it: Use candlesticks as your default; drop to a line chart when you just want to see the clean trend without noise.

What one candle tells you



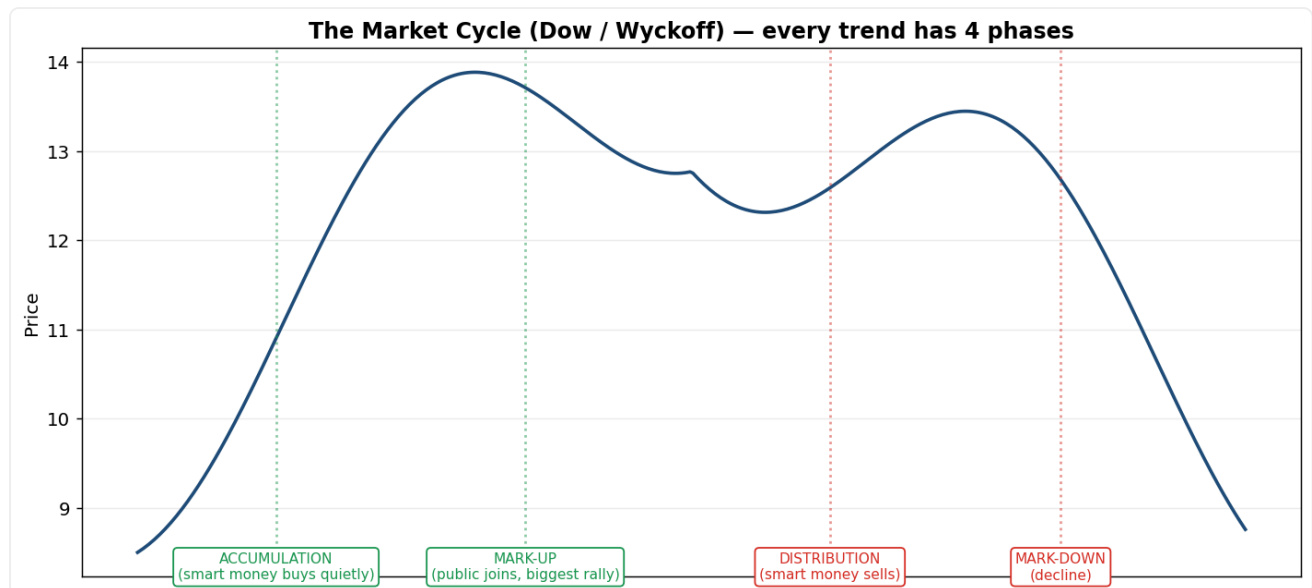
What it is: Body = open-to-close range; wicks/shadows = the high and low; green = closed up, red = closed down.

Why it works: The body shows who won and by how much; the wicks show where price was rejected. A long upper wick means buyers pushed up but sellers slammed it back (supply); a long lower wick means sellers pushed down but buyers absorbed it (demand).

The psychology: A long body reflects conviction — one side overpowered the other all session. A tiny body reflects a stalemate. Wicks are 'rejection': the market visited a price and refused to stay there, which tells you where strong orders sit.

How to use / trade it: Read the wick direction for hidden supply/demand: repeated long lower wicks at a level = buyers defending it (likely support).

The market cycle (Dow / Wyckoff)



What it is: Every trend moves through four phases: Accumulation, Mark-up, Distribution, Mark-down.

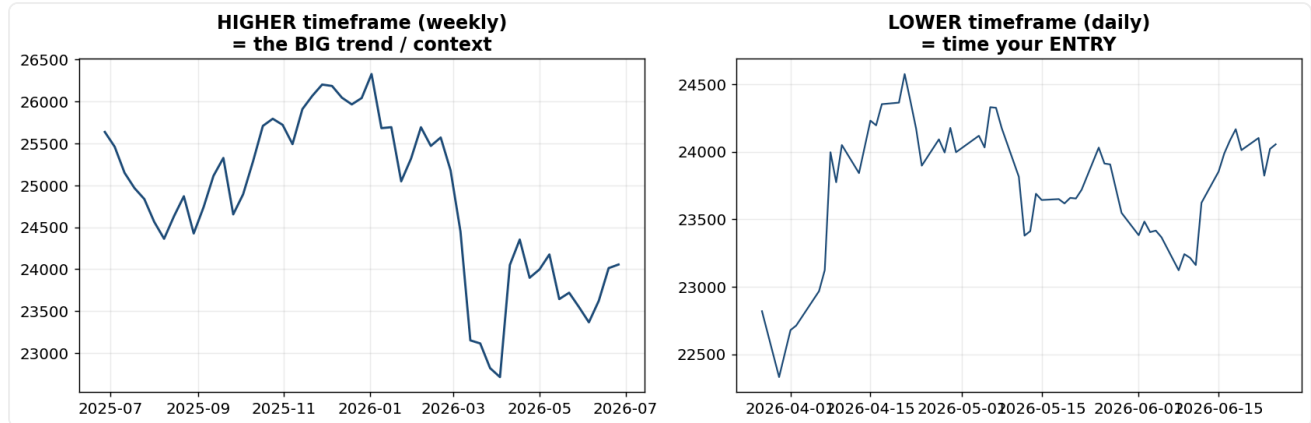
Why it works: Markets are driven by 'smart money' (institutions) who must buy and sell in size without moving price against themselves — so they accumulate quietly at bottoms and distribute quietly at tops, creating the repeating cycle.

The psychology: Accumulation happens during maximum pessimism (everyone has given up selling). Mark-up is when the public notices and FOMO drives the biggest rally. Distribution is peak euphoria, where smart money sells to the late, greedy crowd. Mark-down is the painful unwinding. It's fear → greed → fear.

How to use / trade it: Identify the phase from price+volume: sideways ranges after a big fall (low volume) often = accumulation; sideways after a big rally with high churn = distribution.

Interview depth: This is the backbone of Dow Theory and Wyckoff — name it and explain that smart money buys when retail panics and sells when retail is euphoric.

Multiple-timeframe analysis



What it is: Reading the same asset on a higher timeframe (weekly) and a lower one (daily) together.

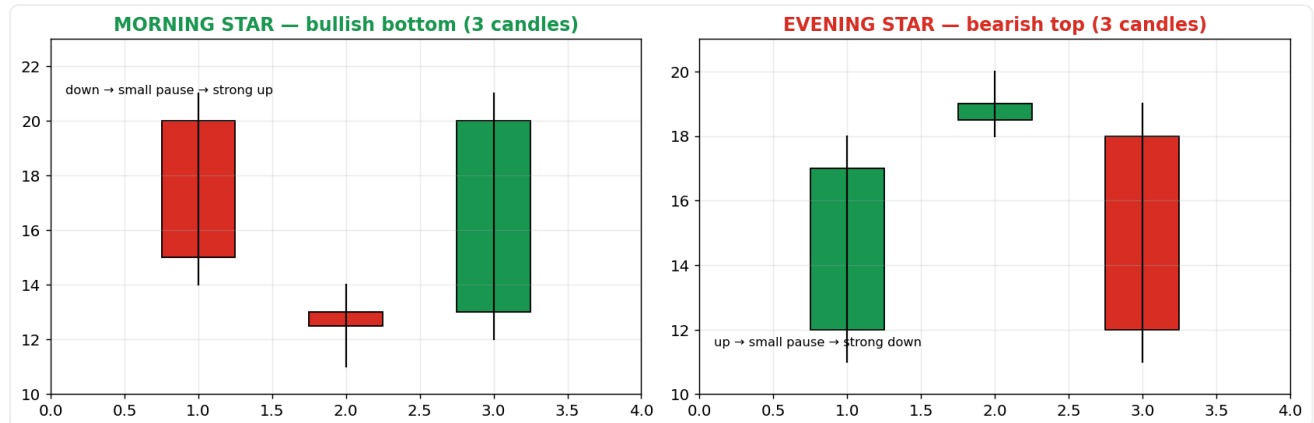
Why it works: A signal aligned with the bigger trend has the wind at its back; a signal against it is fighting a stronger force and usually fails.

The psychology: Different participants act on different timeframes — long-term investors set the tide, short-term traders create the ripples. When both agree, the crowd is unified and moves are powerful.

How to use / trade it: Decide direction on the higher timeframe (only take longs if the weekly trend is up), then time the entry on the lower one. Conflicting timeframes = lower conviction, smaller size or skip.

PART 2 — Price Action & Chart Patterns

Three-candle star reversals



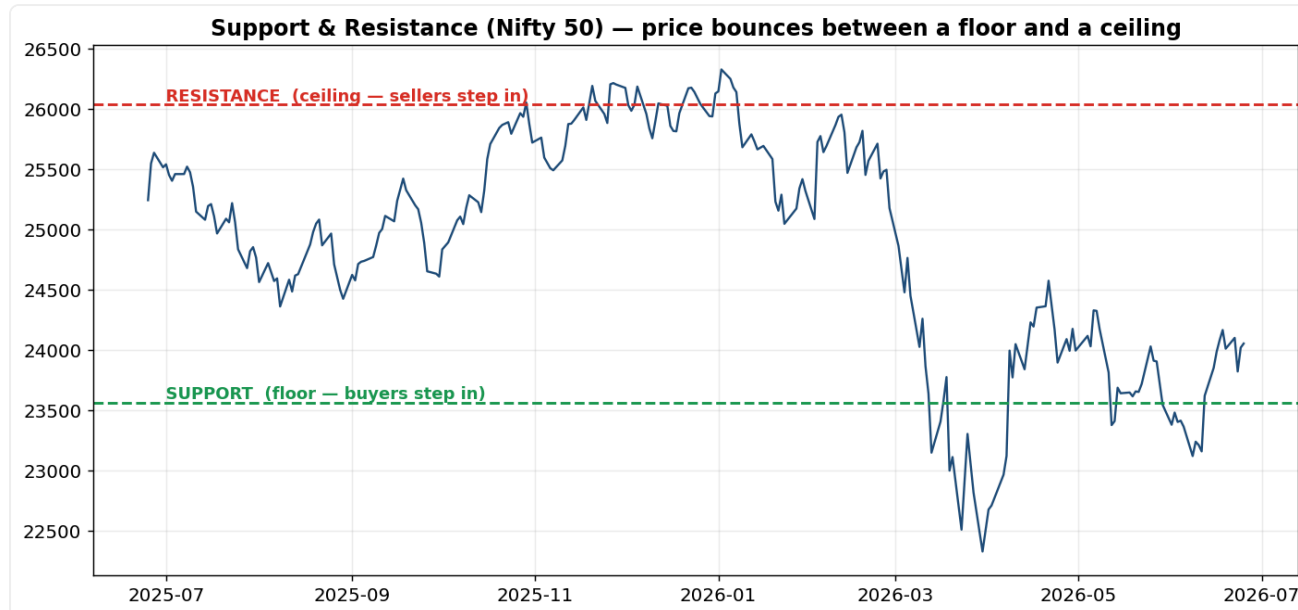
What it is: Morning Star (bullish bottom) and Evening Star (bearish top) — a trend candle, a small 'star' of indecision, then a strong candle the other way.

Why it works: They show a three-act transfer of control: dominance, hesitation, then reversal — more reliable than a single candle because the shift is confirmed over three sessions.

The psychology: Act 1: the trend looks healthy. Act 2: the small star shows the dominant side suddenly can't push further (doubt creeps in). Act 3: the opposite side seizes control with conviction. The pause is the moment the crowd's belief cracks.

How to use / trade it: Enter on the third candle's confirmation; place the stop beyond the star's extreme.

Support & resistance — the floor and the ceiling



What it is: Support = a price floor where buyers repeatedly step in; resistance = a ceiling where sellers repeatedly step in.

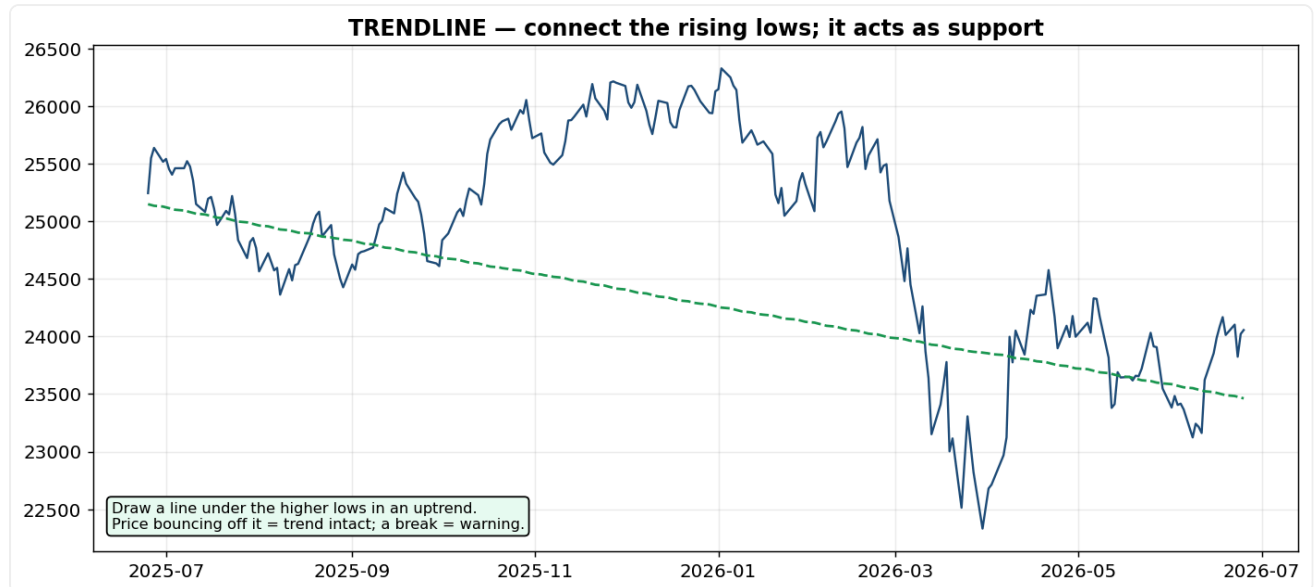
Why it works: They work because of MEMORY and order clustering: traders remember prices where reversals happened and place orders there again, which makes the level hold — partly self-fulfilling.

The psychology: At support three groups buy at once: those who missed the last rally and want a second chance, those who bought there before and add, and shorts taking profit. That clustered demand stops the fall. **Role reversal:** when resistance finally breaks, traders who sold there are now trapped at a loss and buy back at breakeven on the retest — turning old resistance into new support.

How to use / trade it: Buy near support / sell near resistance with a stop just beyond the level; trade the breakout when price clears it on strong volume.

Interview depth: The more times a level is tested the more significant it is — but each test consumes orders, so a level tested many times eventually breaks (the supply/demand there gets used up).

Trendlines



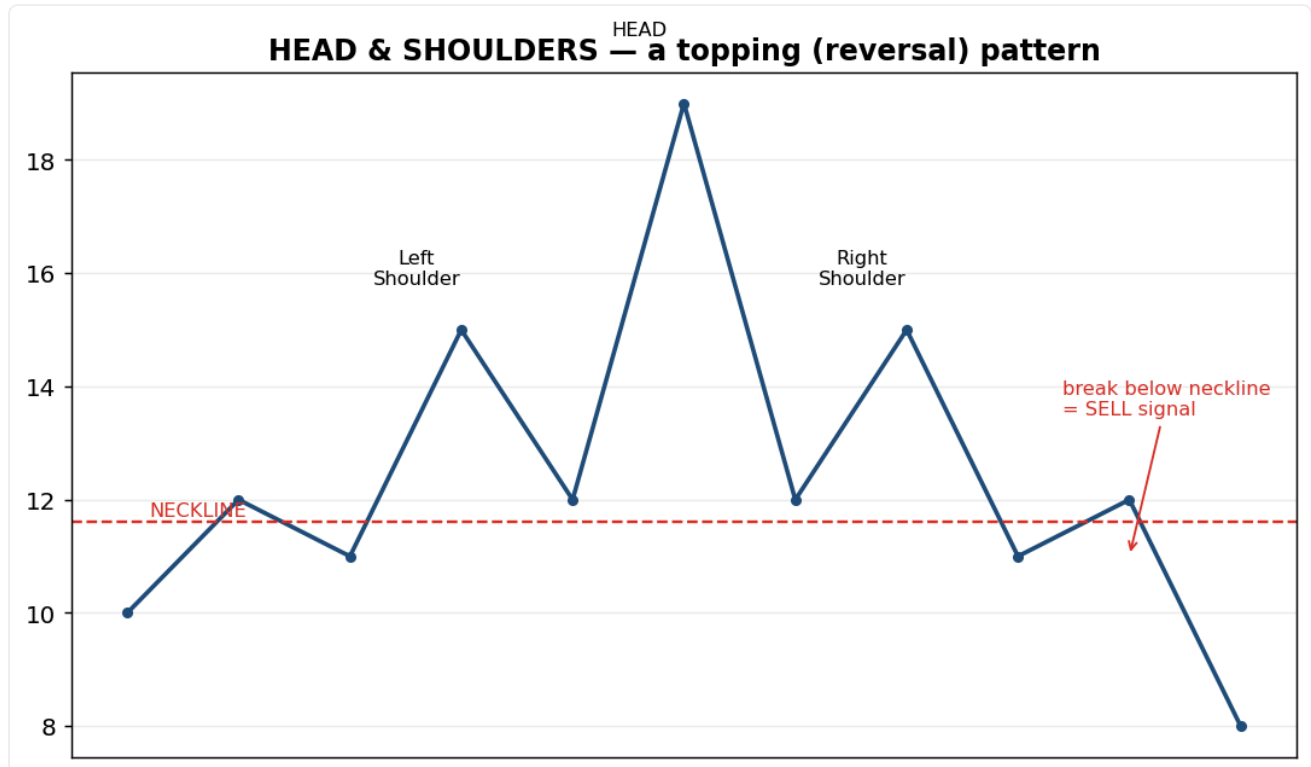
What it is: A line connecting the rising lows of an uptrend (or falling highs of a downtrend); a sloping support/resistance.

Why it works: It visualises that buyers are consistently willing to step in at HIGHER prices each time — proof of increasing demand.

The psychology: Each bounce off the line is buyers showing growing optimism (paying up earlier each dip). A break of the line means that rising demand has finally failed — the psychological 'agreement' to keep buying higher has broken.

How to use / trade it: Needs two touches to draw, a third to confirm. Buy bounces off the line in an uptrend; treat a decisive break as an early warning to tighten stops.

Head & Shoulders (top reversal)



What it is: Three peaks — a higher middle peak (head) between two lower peaks (shoulders) — with a 'neckline' support; a break below it signals a top.

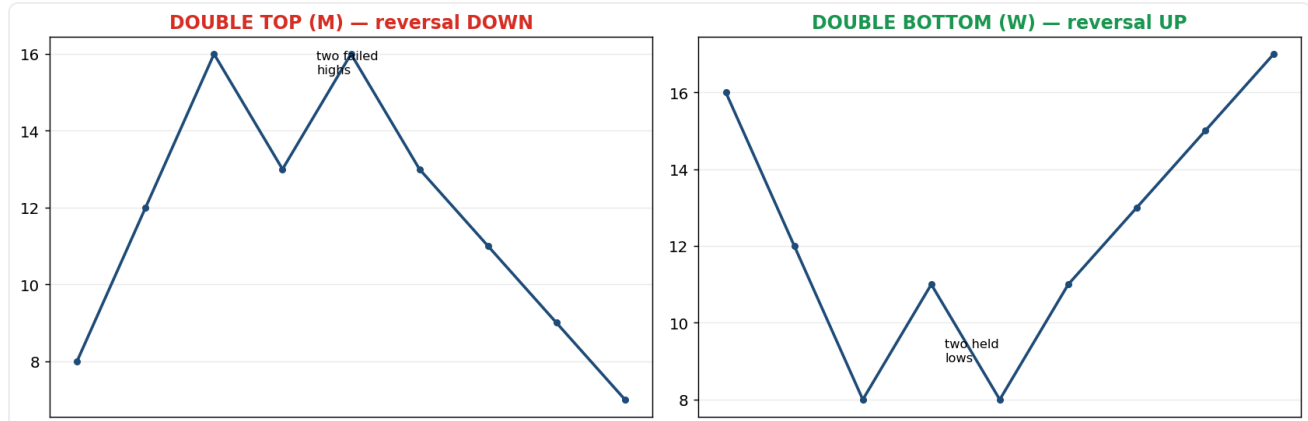
Why it works: It maps a trend visibly running out of buyers: each rally attempt has less force, and the final support break unleashes trapped longs.

The psychology: Left shoulder = a normal rally with healthy volume. Head = a euphoric push to a new high but usually on LOWER volume — the last greedy buyers. Right shoulder = buyers try again but can't even make a new high — demand is exhausted. When the neckline breaks, everyone who bought the top is underwater and sells, accelerating the fall.

How to use / trade it: Sell on the neckline break (ideally a retest of it from below); target $\approx$ the head-to-neckline distance projected down. Volume should fall through the head and spike on the break.

Interview depth: The volume signature is the tell — point out that lower volume on the head vs the left shoulder reveals weakening conviction even before the break.

Double top & double bottom



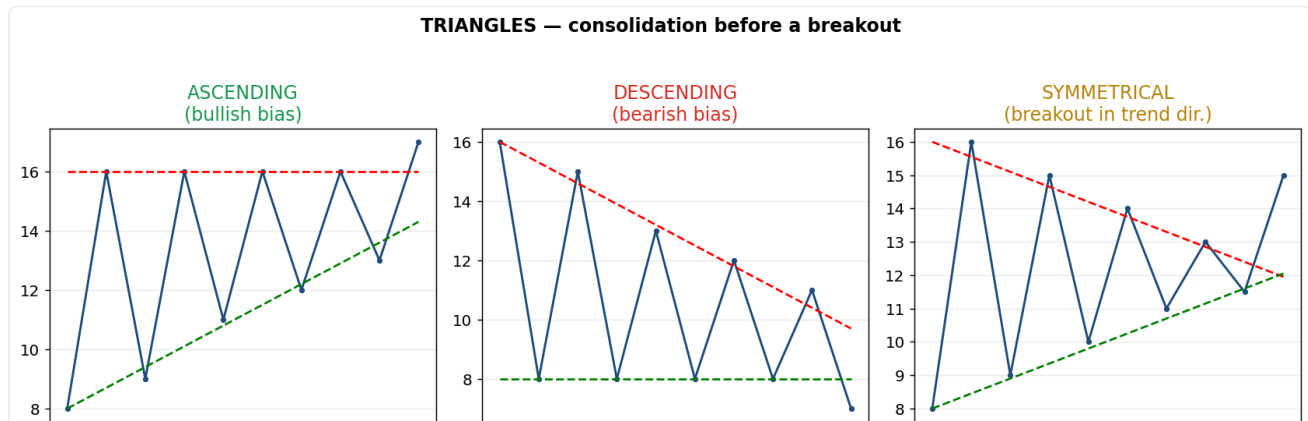
What it is: Two failed attempts at the same level — M-shape (top, reversal down) or W-shape (bottom, reversal up).

Why it works: A level that rejects price twice proves strong opposing orders sit there; the second failure convinces the trend-followers to give up.

The psychology: At a double top, buyers test the high, fail, and the second failure signals that demand can't absorb the supply at that price. Breakout buyers who bought the second push get trapped and bail, fuelling the reversal. The mirror logic applies to a double bottom.

How to use / trade it: Enter when price breaks the 'neckline' (the middle low of an M / middle high of a W); target $\approx$ the pattern's height.

Triangles (consolidation)



What it is: Price coils into a narrowing range — ascending (flat top, rising lows), descending (flat bottom, falling highs), or symmetrical (both converging).

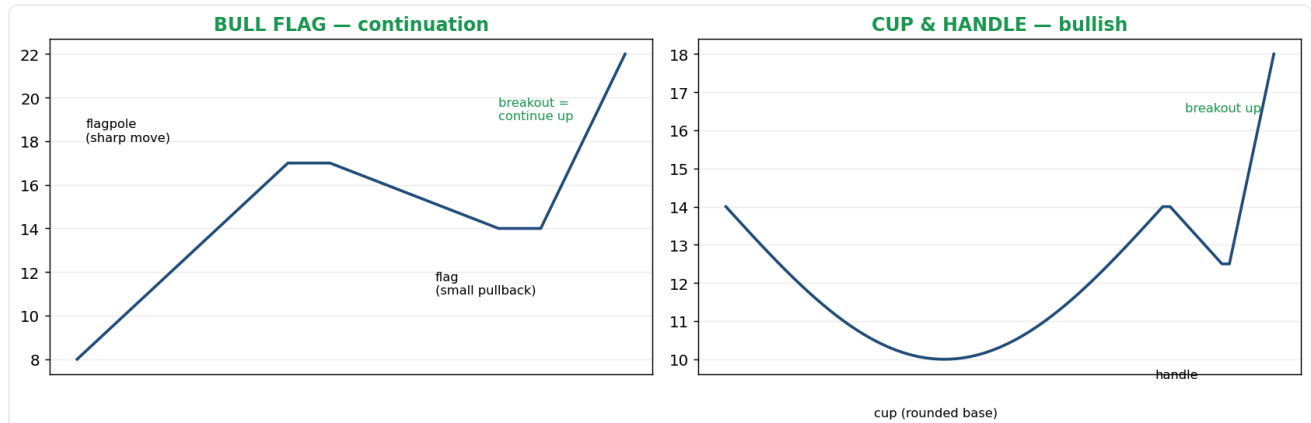
Why it works: A triangle is a tightening tug-of-war; the breakout direction reveals which side won the accumulation of orders.

The psychology: Ascending: buyers keep paying higher lows (rising demand) while a wall of sellers caps a level — eventually buyers absorb all that supply and price breaks up. **Descending:** the reverse, sellers grind buyers down.

Symmetrical: both sides compress; the breakout shows whose conviction was real. Volatility contracts (the 'calm') then expands violently on the break.

How to use / trade it: Trade the breakout in the direction of the break, ideally with a volume surge; the prior trend usually wins in a symmetrical triangle.

Flags & Cup-and-Handle (continuation)



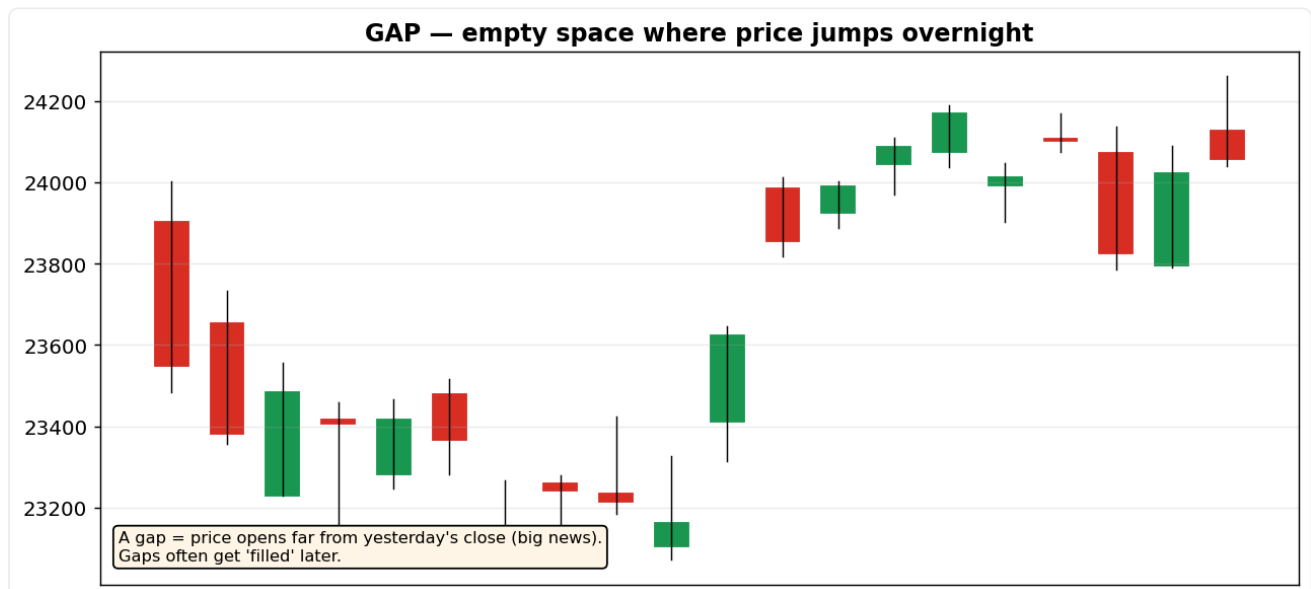
What it is: A flag is a brief counter-trend pullback after a sharp move (the 'flagpole'); a cup & handle is a rounded base then a small dip before breaking out.

Why it works: Strong moves rarely reverse instantly — they pause to let profit-takers exit and new buyers position, then resume.

The psychology: After a sharp rally, early buyers take profit (the pullback), which shakes out weak/nervous hands. But the dominant side is still in control, so once the weak holders are flushed, the trend continues. The cup's rounded shape is the slow shift from despair to hope; the handle is the final shakeout before the breakout.

How to use / trade it: Enter on the breakout from the flag/handle; target $\approx$ the flagpole height added to the breakout point.

Gaps



What it is: Empty space on the chart where price opens far from the previous close — usually on overnight news.

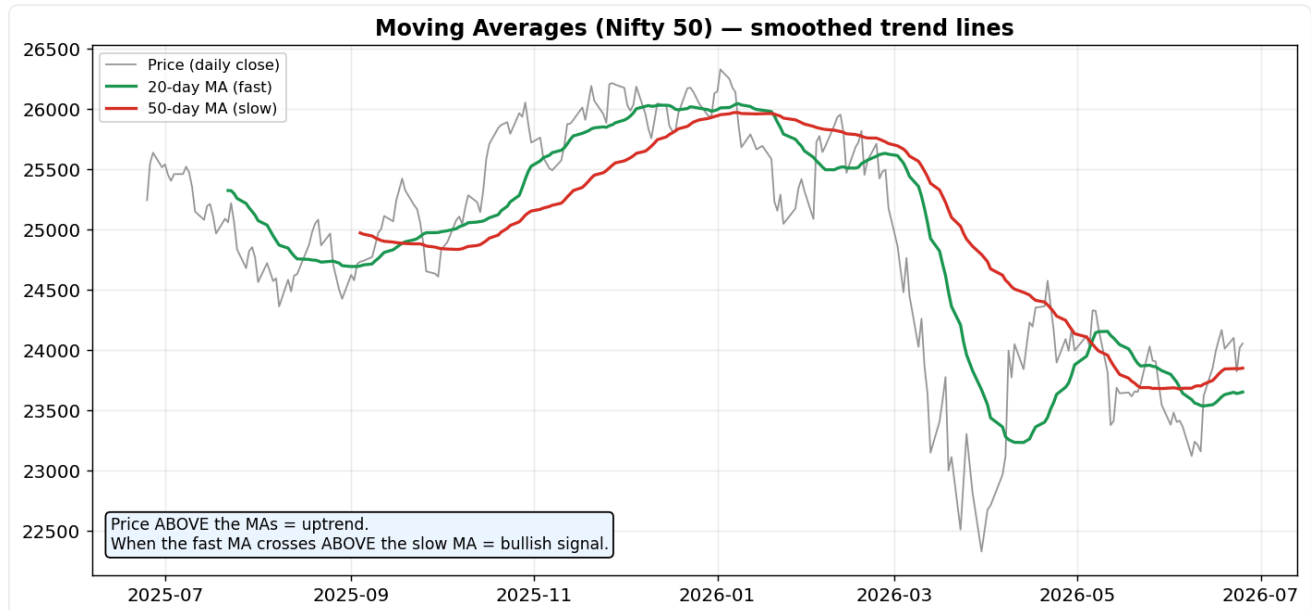
Why it works: A gap marks a sudden supply/demand imbalance strong enough that no trading happened in between.

The psychology: Gaps are pure emotion crystallised: a breakaway gap = new information genuinely repricing the asset; an exhaustion gap = the last euphoric buyers or panicked sellers piling in at the end of a move. Many gaps 'fill' later because the initial reaction was an emotional overshoot that rational traders fade.

How to use / trade it: Breakaway gaps (with volume) often start strong trends — trade with them; exhaustion gaps often mark ends — fade them. Watch for the gap to fill as a target.

PART 3 — Indicators (and why each works)

Moving averages



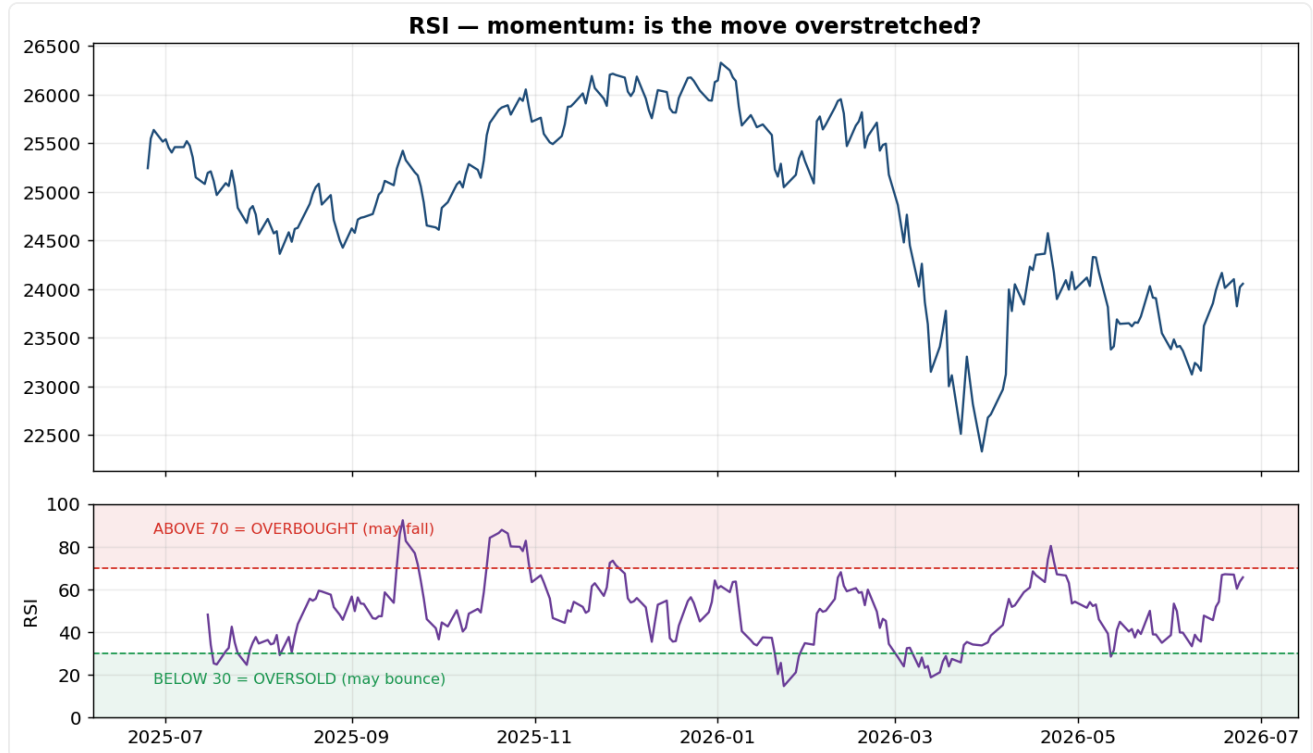
What it is: The average closing price over the last N days, re-plotted daily, smoothing out noise.

Why it works: It represents the average price participants paid recently — a moving 'consensus value' that the market gravitates around.

The psychology: When price is above the MA, the average holder is in profit and feels confident (bullish bias); below it, holders are underwater and nervous. Traders treat the MA as a fair-value reference and buy dips to it, which is why it acts as dynamic support. A golden cross (fast above slow) means the recent average cost has risen above the long-term — sentiment has structurally turned up.

How to use / trade it: Use price-vs-MA for trend bias and MA crossovers for shifts; the 20/50/200-day are the most-watched (self-fulfilling because everyone uses them).

RSI — overbought / oversold



What it is: A 0–100 oscillator measuring the strength of recent gains versus losses (default 14 periods).

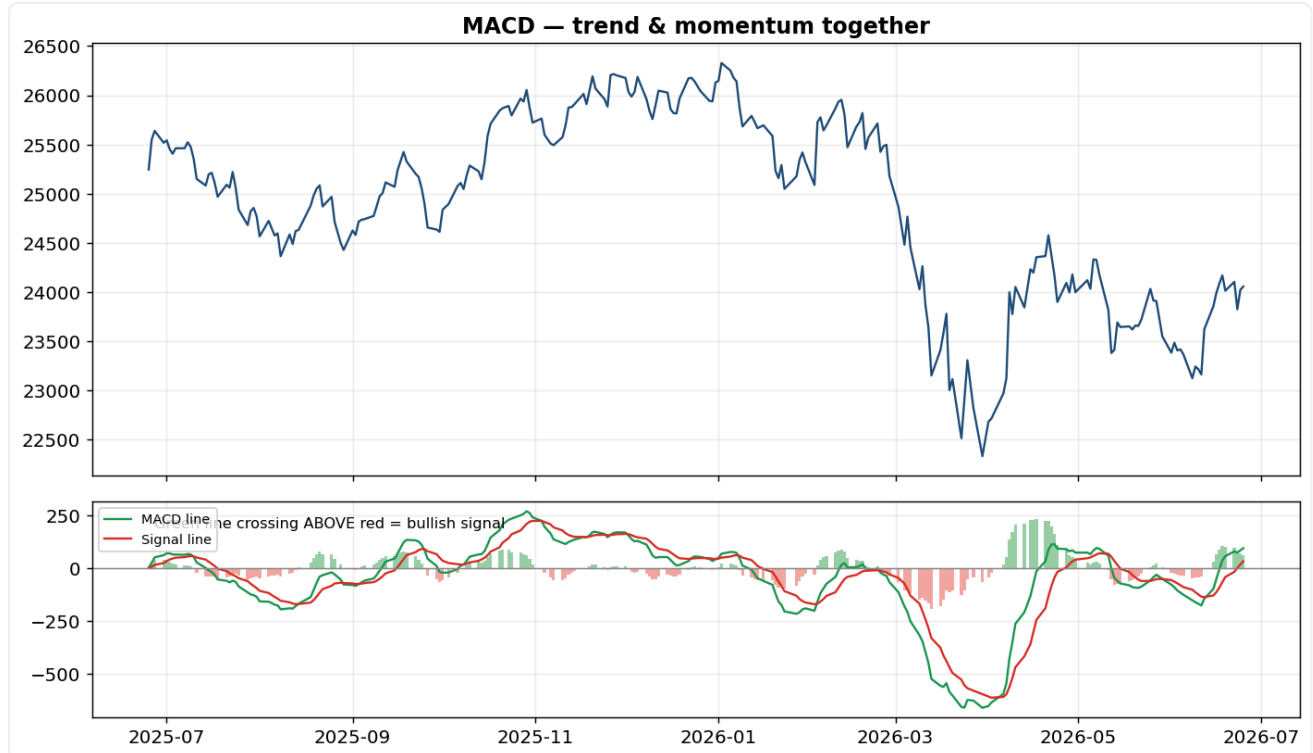
Why it works: It quantifies momentum — how one-sided the buying or selling has been — so you can spot when a move is overstretched.

The psychology: Above 70 (overbought) means buying has been so aggressive that nearly everyone who wanted in is already in — there's little fuel left, so a pause/pullback is likely. Below 30 (oversold) means panic selling has likely exhausted itself. **Divergence** is the deeper signal: if price makes a new high but RSI doesn't, the rally is being driven by fewer and weaker buyers — momentum is dying.

How to use / trade it: Use it as a warning light, not a trigger; in strong trends RSI can stay extreme for a long time, so combine it with price action.

Interview depth: Emphasise that 'overbought' ≠ 'sell now' — in a powerful uptrend shorting overbought RSI is a classic beginner mistake; divergence is the higher-quality signal.

MACD — trend + momentum



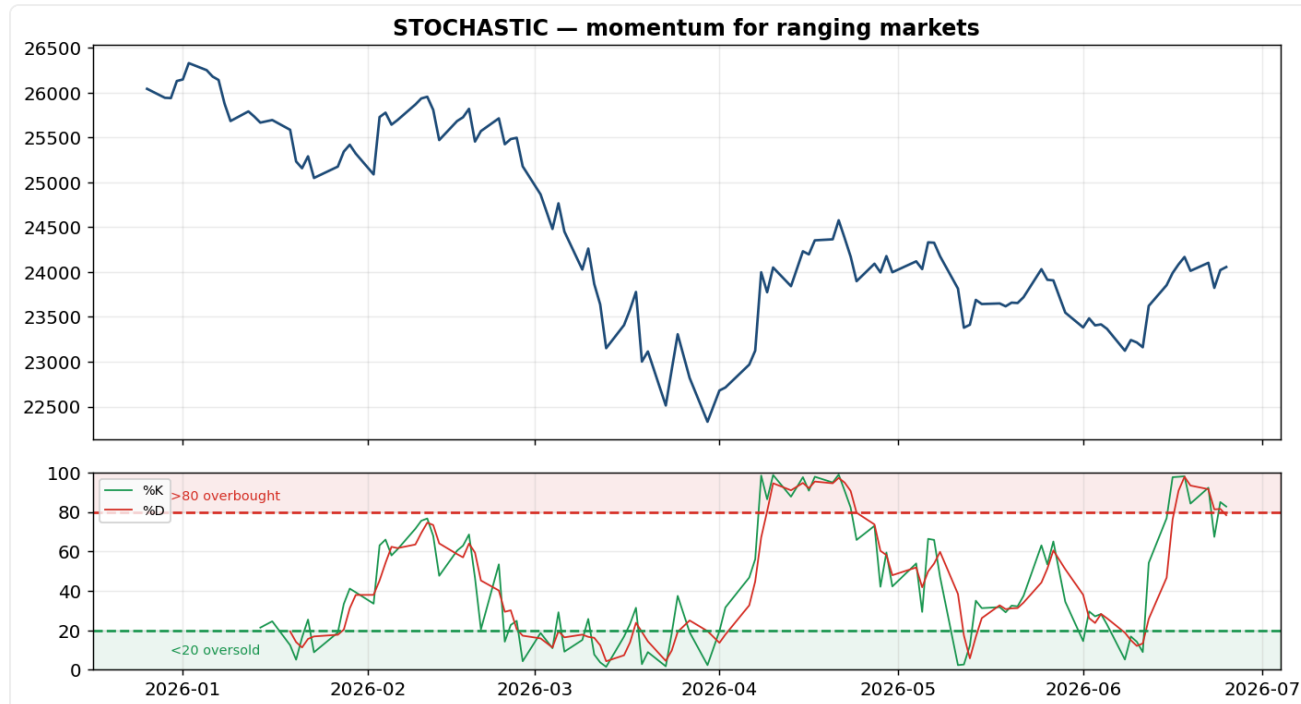
What it is: The gap between a fast (12) and slow (26) EMA, with a 9-EMA signal line and a histogram of the gap.

Why it works: It captures the rate at which the short-term average is pulling away from the long-term — i.e., accelerating or fading momentum.

The psychology: A bullish crossover means short-term buying pressure has overtaken the longer-term trend — the balance of power is shifting to buyers in real time. The histogram shrinking even while price rises warns that momentum is fading before price actually turns.

How to use / trade it: Use crossovers for entries/exits and histogram + divergence for early warnings; best as confirmation alongside trend and levels.

Stochastic oscillator



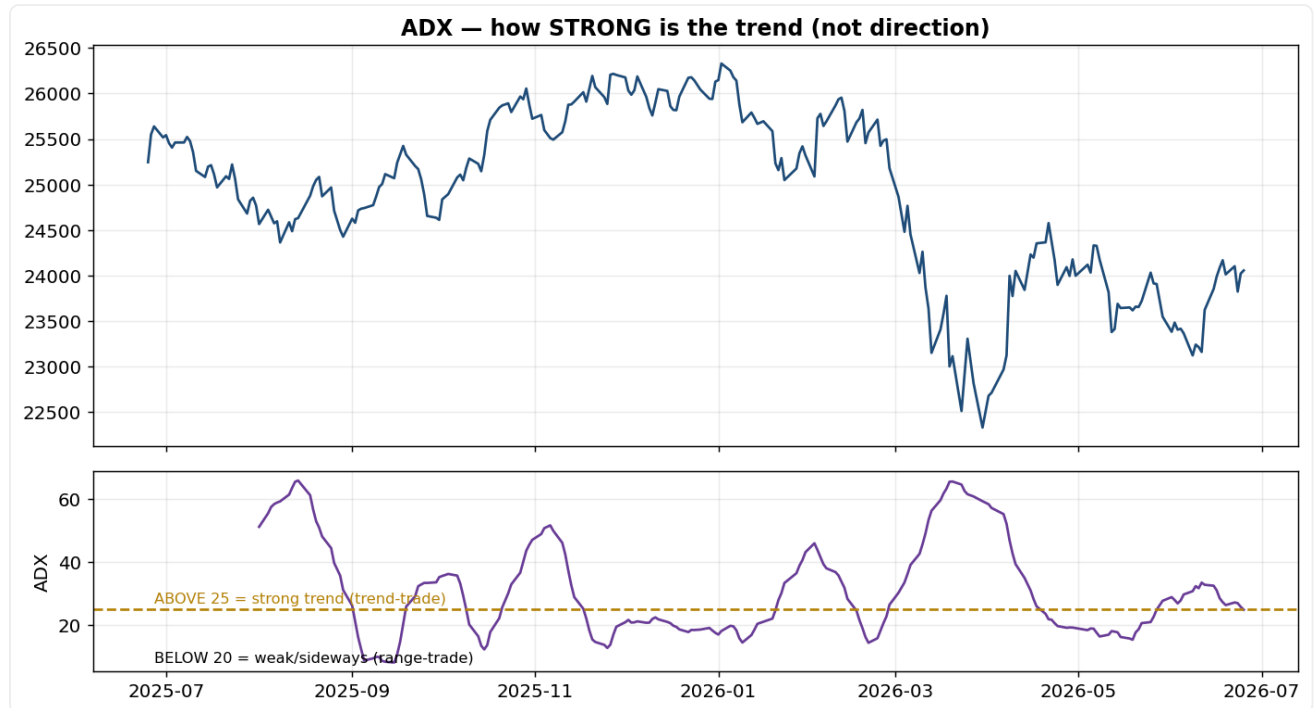
What it is: Shows where the close sits within the recent high-low range (0–100), with %K and %D lines.

Why it works: Closing near the top of the range signals strength; near the bottom, weakness — a quick read on momentum, especially in ranges.

The psychology: In an uptrend price tends to close near its highs (confident buyers); when it starts closing in the lower part of the range despite higher prices, buyers are tiring. Overbought/oversold zones flag emotional extremes in range-bound markets.

How to use / trade it: Best in sideways markets; use %K/%D crossovers in the 80/20 zones, and watch for divergence.

ADX — trend STRENGTH



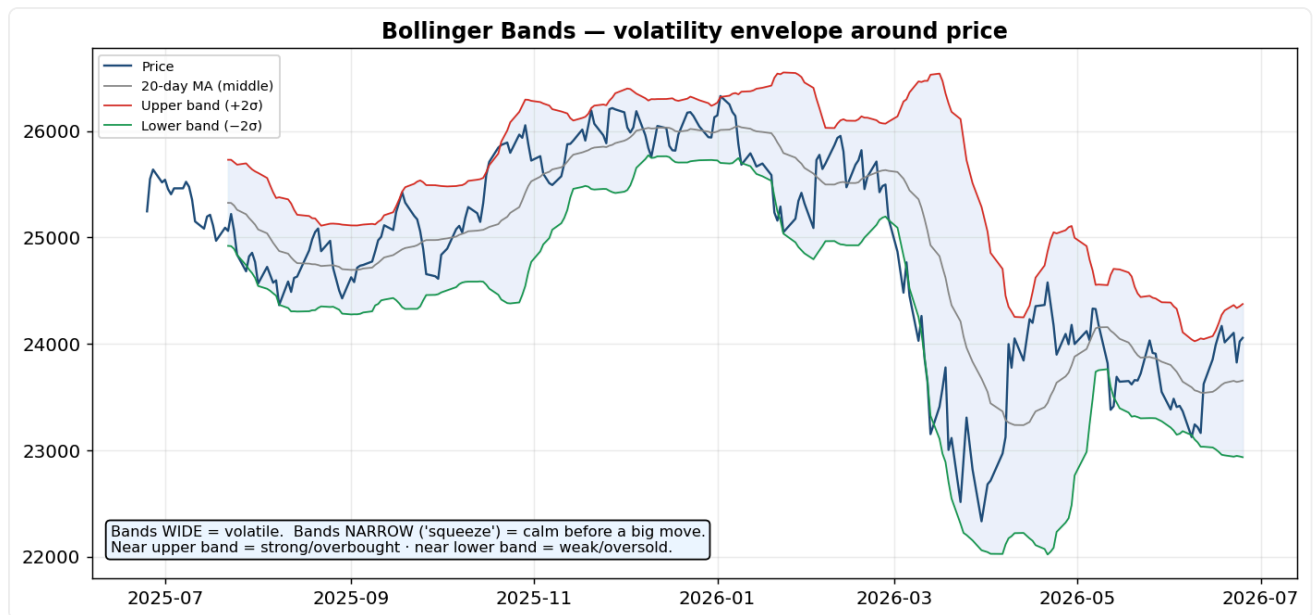
What it is: A 0–100 line measuring how strong a trend is, regardless of direction.

Why it works: Many tools only work in the right regime — ADX tells you whether to use trend tools or range tools.

The psychology: A high ADX means participants broadly agree on direction (strong conviction, persistent trend); a low ADX means disagreement/indecision (choppy range). It's a measure of crowd consensus strength.

How to use / trade it: Above 25 → trade with the trend (let winners run); below 20 → switch to range tactics (fade extremes). Don't use RSI overbought-as-sell when ADX is high — the trend will keep pushing.

Bollinger Bands — volatility



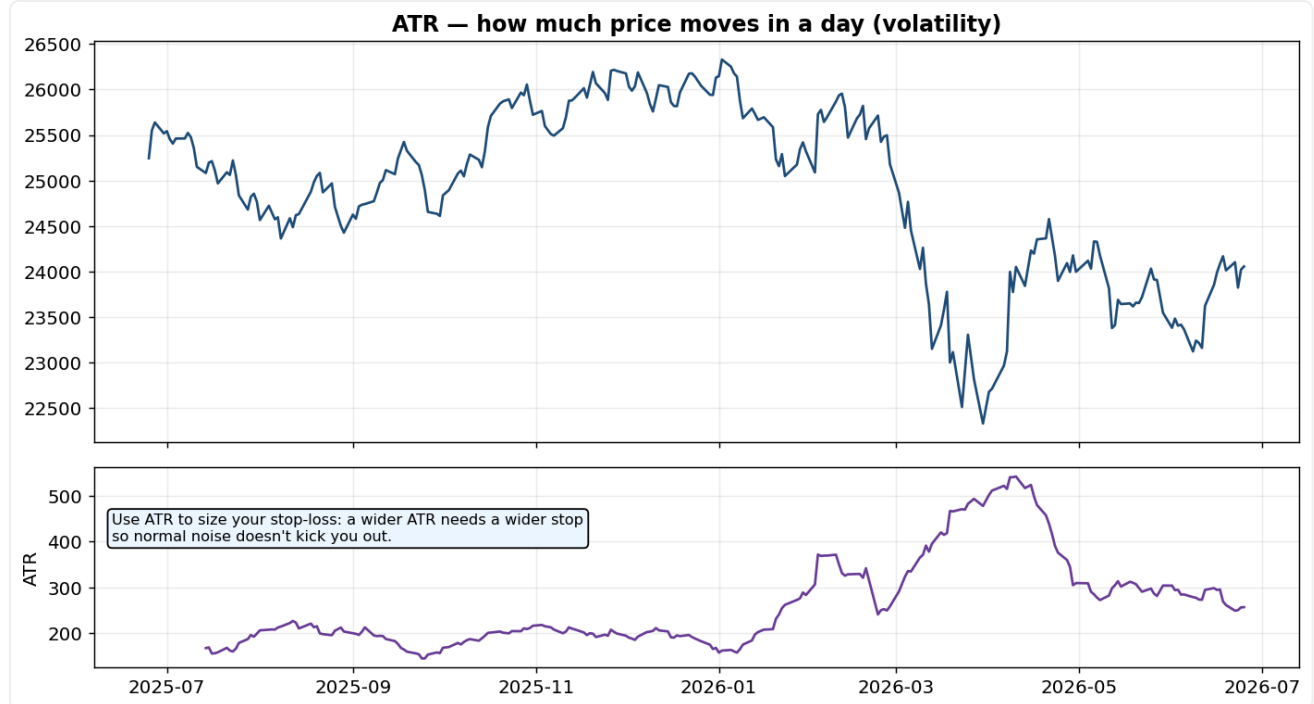
What it is: A 20-day average with bands set 2 standard deviations above and below it.

Why it works: Two standard deviations statistically contain ~95% of recent prices, so touches of the bands flag stretched conditions, and band WIDTH measures volatility.

The psychology: A 'squeeze' (very narrow bands) reflects low disagreement and complacency — a coiled spring; when new information arrives, the pent-up move releases violently (expansion). Price hugging the upper band shows persistent strength, not just 'overbought'.

How to use / trade it: Trade squeeze breakouts in the breakout direction; in a range, fade band touches back to the middle average.

ATR — how much it moves



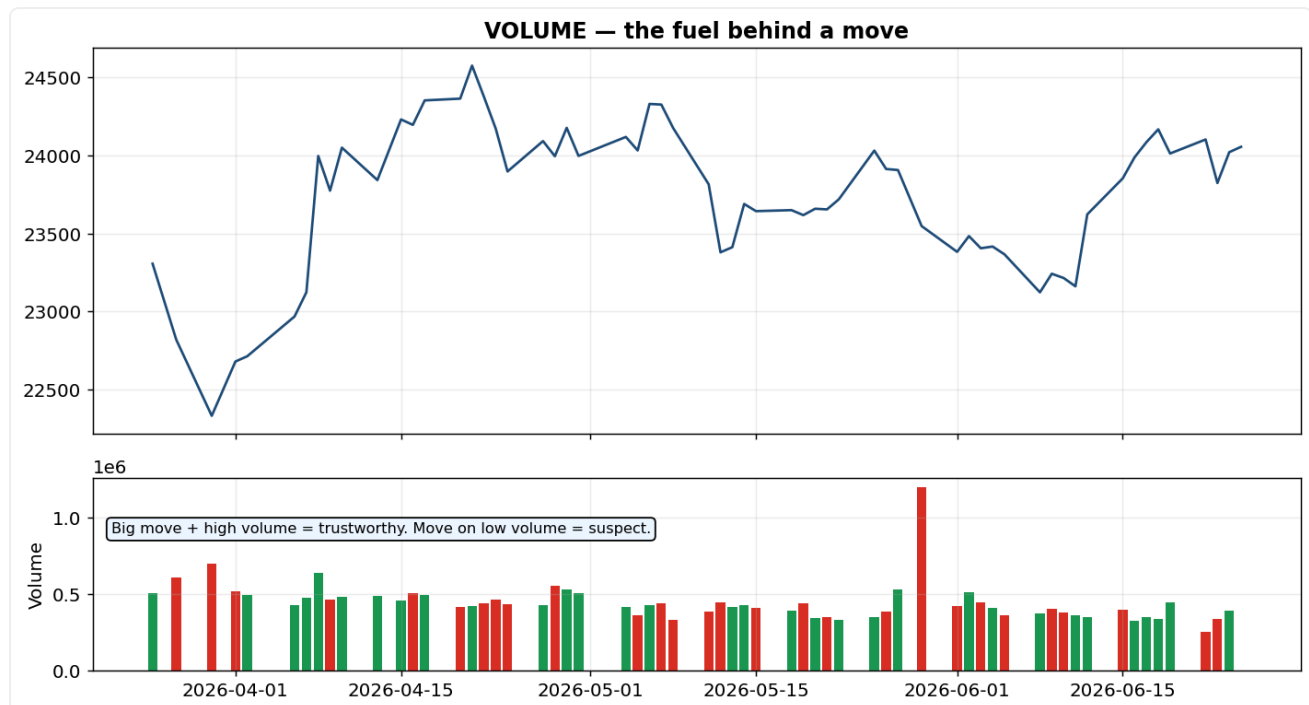
What it is: Average True Range — the typical size of a day's move, including gaps (pure volatility, no direction).

Why it works: Risk must be scaled to volatility; ATR gives you that yardstick objectively.

The psychology: Volatility is emotional intensity — ATR rises when fear/excitement spikes. A stop placed without regard to ATR will be too tight in a volatile name (stopped out by noise) or too loose in a calm one.

How to use / trade it: Set stops at $\sim 1.5\text{--}2\times$ ATR from entry and size positions so each trade risks a similar rupee amount regardless of the asset's volatility.

Volume — the fuel



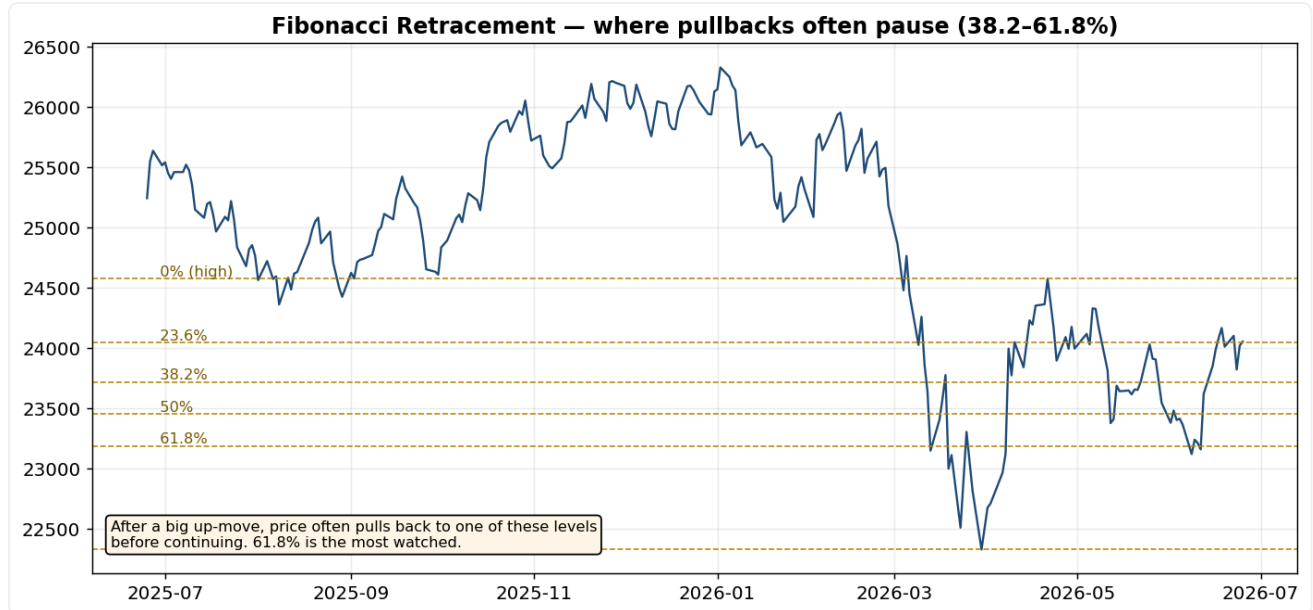
What it is: The number of shares/contracts traded; shown as bars under price.

Why it works: Price tells you the direction; volume tells you the conviction behind it.

The psychology: A big move on heavy volume means many participants agree — real money is committed, so it's likely to continue. The same move on thin volume means few believe it; it's fragile and often reverses. Rising volume into a breakout confirms genuine demand.

How to use / trade it: Demand volume confirmation on breakouts; be sceptical of moves and patterns that complete on weak volume.

Fibonacci retracement



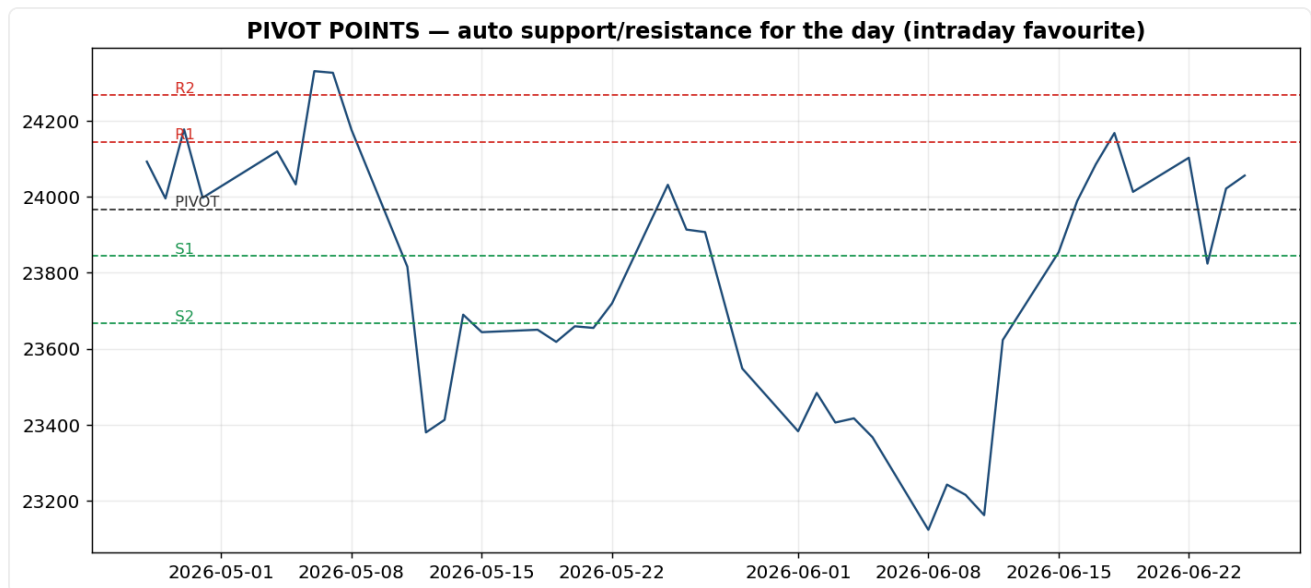
What it is: Horizontal levels at 23.6%, 38.2%, 50% and 61.8% of a prior move, marking likely pullback zones.

Why it works: It works through a mix of natural profit-taking rhythm and self-fulfilment — so many traders watch these levels that orders cluster there.

The psychology: After a rally, some buyers take profit, causing a dip. The 38.2–61.8% zone is where the pullback has given back enough to shake out weak hands and tempt new buyers who missed the move — so demand re-emerges there. A 61.8% retracement is the deepest 'healthy' pullback before the trend is in doubt.

How to use / trade it: In an uptrend, buy dips into the 38–62% zone, especially where a Fib level lines up with support or a moving average (confluence = higher probability).

Pivot points



What it is: Support (S1, S2) and resistance (R1, R2) levels auto-calculated from the prior day's high, low and close, around a central pivot.

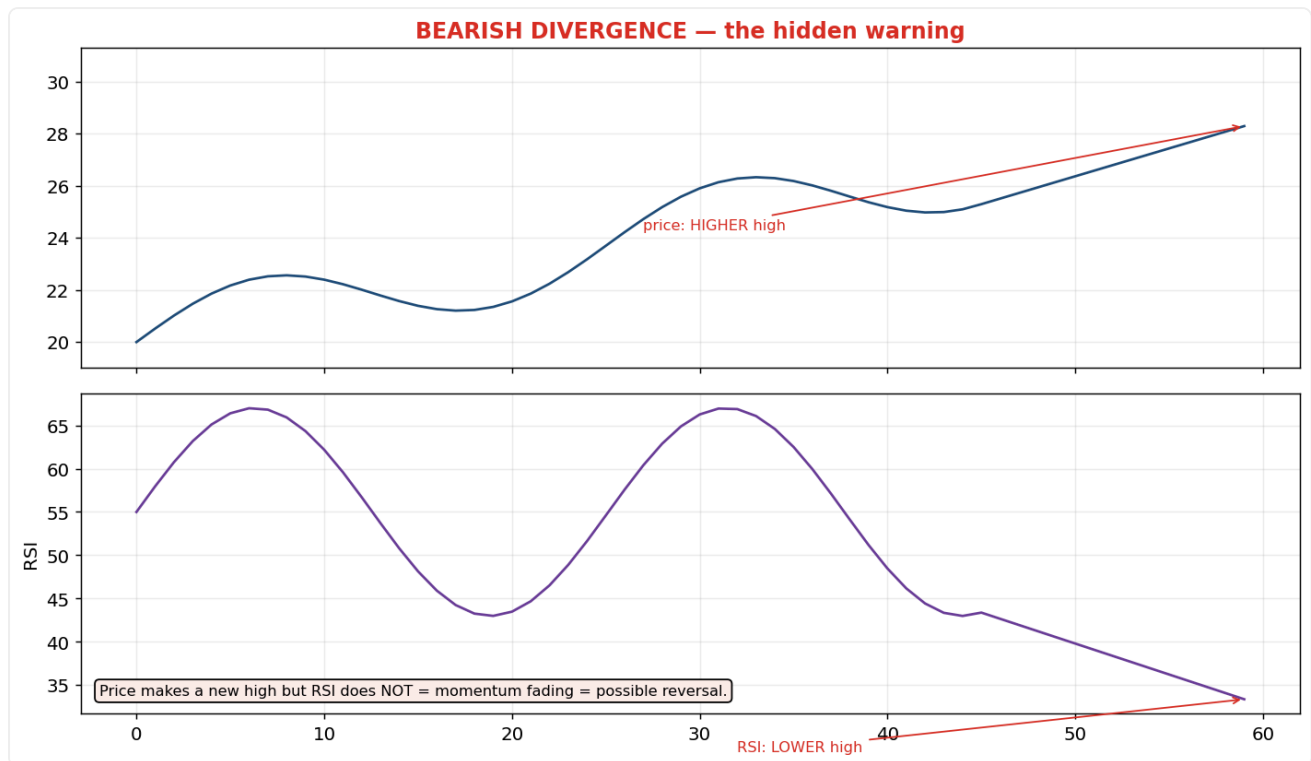
Why it works: They give objective intraday reference levels derived from where the market just agreed value was.

The psychology: Because day-traders and algos all compute the same pivots, orders cluster at them — they become self-fulfilling intraday magnets and barriers. Price above the central pivot = bullish bias for the session.

How to use / trade it: Use them for intraday targets and stops on indices and F&O; trade bounces/rejections at S/R levels and breakouts through them.

PART 4 — Theories

Divergence — the hidden warning



What it is: Price and a momentum indicator (RSI/MACD) disagree — e.g., price makes a higher high but the indicator makes a lower high.

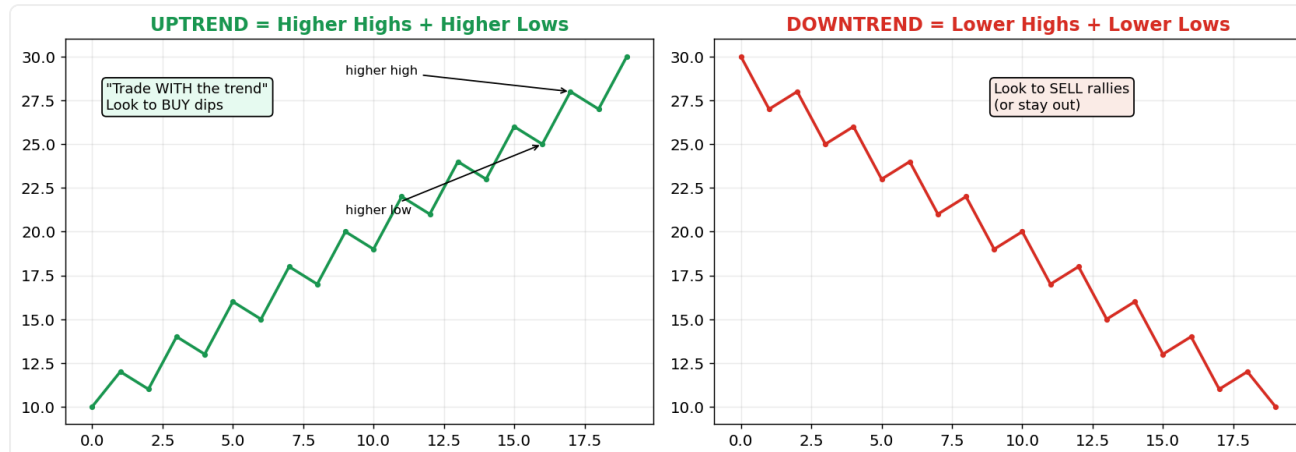
Why it works: Price can keep rising on diminishing participation; the indicator exposes that the engine is weakening before the car stops.

The psychology: A new price high on weaker momentum means the rally is being carried by fewer, later, and smaller buyers — often smart money is quietly distributing to an euphoric crowd. The crowd sees a new high and feels safe; the divergence reveals the opposite.

How to use / trade it: Treat divergence as an early-warning to tighten stops or look for a reversal trigger — not a stand-alone sell; wait for price confirmation (a break of support).

Interview depth: Distinguish regular divergence (warns of reversal) from hidden divergence (signals trend continuation) — mentioning both shows real depth.

Trend structure



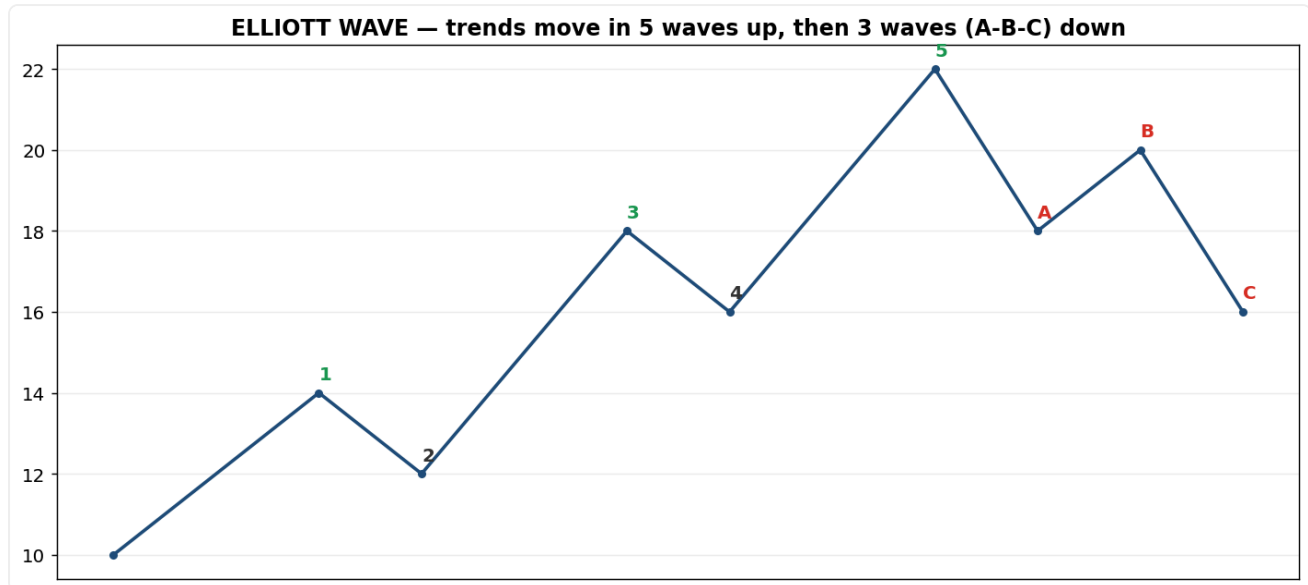
What it is: Uptrend = higher highs and higher lows; downtrend = lower highs and lower lows; sideways = a range.

Why it works: Trends persist because of feedback loops — rising prices attract more buyers and validate holders, which pushes prices higher still.

The psychology: An uptrend is a visible story of growing optimism: each dip is bought earlier (higher lows = impatient demand) and each push reaches further (higher highs). The trend ends when that optimism breaks — the first lower high/low signals the psychology has shifted.

How to use / trade it: Always define the trend first and trade with it: buy dips in uptrends, sell rallies in downtrends, stand aside in choppy ranges.

Elliott Wave



What it is: Trends unfold in 5 waves in the trend direction (1-3-5 up, 2-4 pullbacks), then correct in 3 waves (A-B-C).

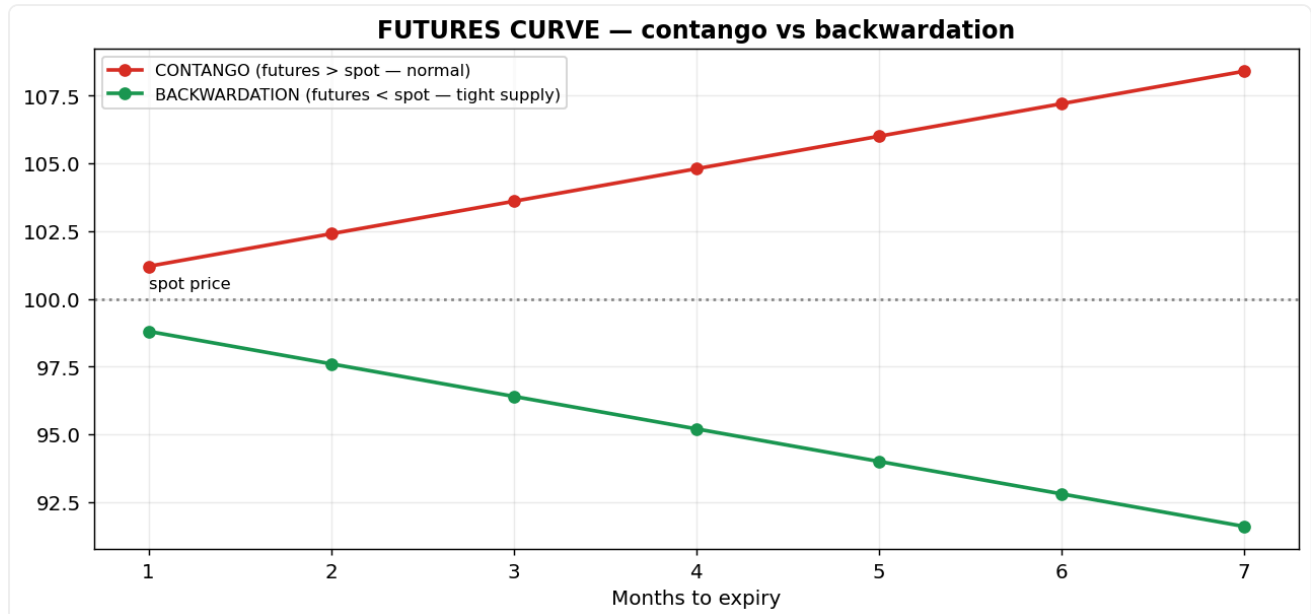
Why it works: It formalises the natural swing of crowd emotion between optimism and pessimism, and is fractal (the same shape on every timeframe).

The psychology: Wave 1 = a few early believers. Wave 2 = doubt (pullback). Wave 3 = the crowd recognises the trend — the strongest, longest wave (peak optimism building). Wave 4 = profit-taking. Wave 5 = euphoria/exhaustion, often on divergence. Then A-B-C unwinds the excess as reality sets in.

How to use / trade it: Use it to gauge where a move is in its life-cycle — e.g., be cautious chasing a 'wave 5' that shows momentum divergence. Acknowledge it's subjective and best combined with other tools.

PART 5 — Futures & Options (F&O)

Futures: contango vs backwardation



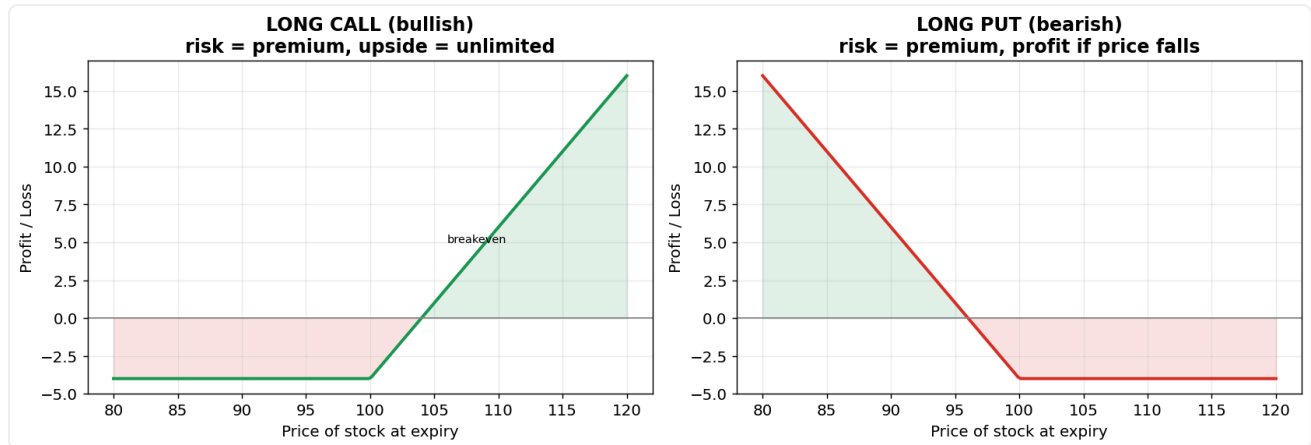
What it is: The shape of futures prices across expiries — contango (futures above spot) or backwardation (futures below spot).

Why it works: The curve reflects the cost of carrying the asset versus how urgently the market wants it now.

The psychology: Contango is 'normal' — futures cost more because of interest/storage (cost of carry). Backwardation signals near-term scarcity or strong immediate demand — buyers will pay a premium for the asset NOW rather than later (common in tight crude/commodity markets), a sign of bullish urgency.

How to use / trade it: Watch the curve and rollovers: heavy long rollover to the next expiry signals trend conviction; backwardation often accompanies strong spot demand.

Calls & Puts (the basics)



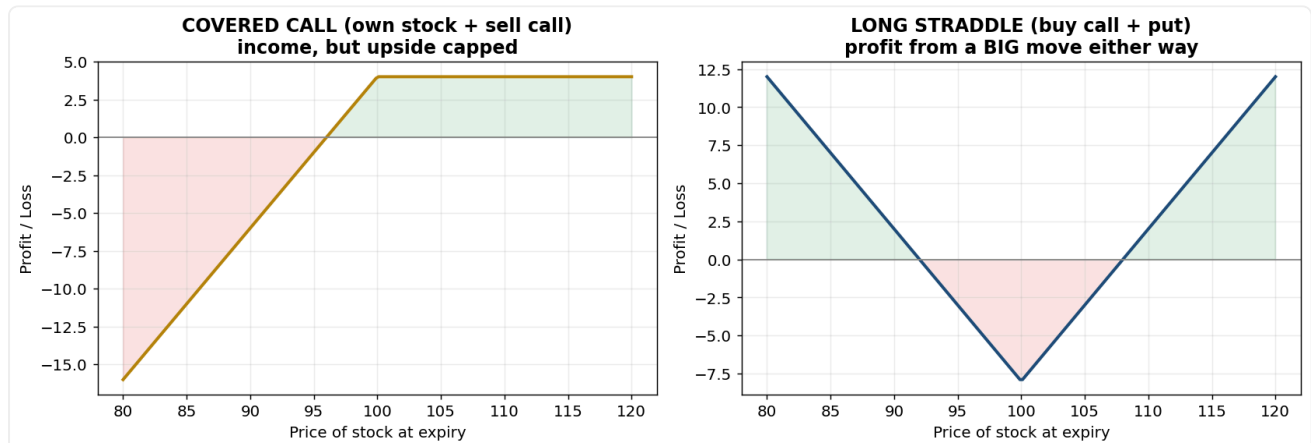
What it is: A call is the right to buy at a strike; a put is the right to sell. Buyers pay a premium and risk only that premium.

Why it works: Options let you take a directional or volatility view with defined, limited risk and built-in leverage.

The psychology: Option buyers pay a small premium for large potential upside (limited risk, asymmetric payoff) — attractive to speculators. Option sellers (writers) collect that premium and profit from time decay, betting the move won't happen — they're effectively the 'insurance company'.

How to use / trade it: Buy calls when bullish, puts when bearish; remember time decay (theta) works against buyers every day, so timing matters.

Covered call & straddle



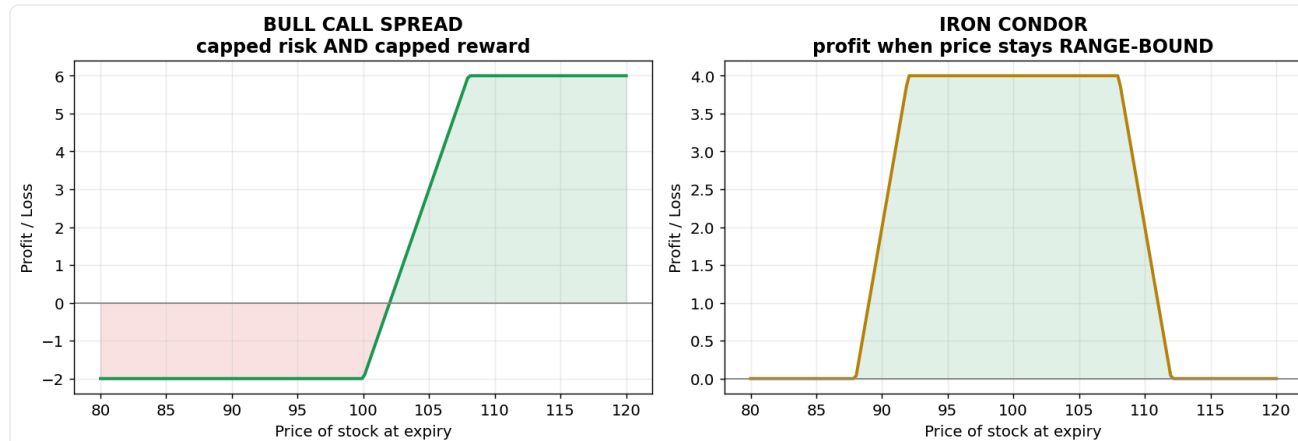
What it is: Covered call = own the stock and sell a call for income. Long straddle = buy a call AND a put at the same strike.

Why it works: These shape your payoff to a specific view: income in a flat market, or a bet on a big move either way.

The psychology: A covered-call seller is willing to cap upside in exchange for premium income — a mildly bullish/neutral, income-seeking mindset. A straddle buyer doesn't care about direction, only that volatility will be large — typically bought before events (results, budgets) when a big move is expected but the direction is unknown.

How to use / trade it: Covered calls for income on holdings; straddles/strangles ahead of high-uncertainty events — but they need a move bigger than the combined premium to pay off.

Spreads & Iron Condor



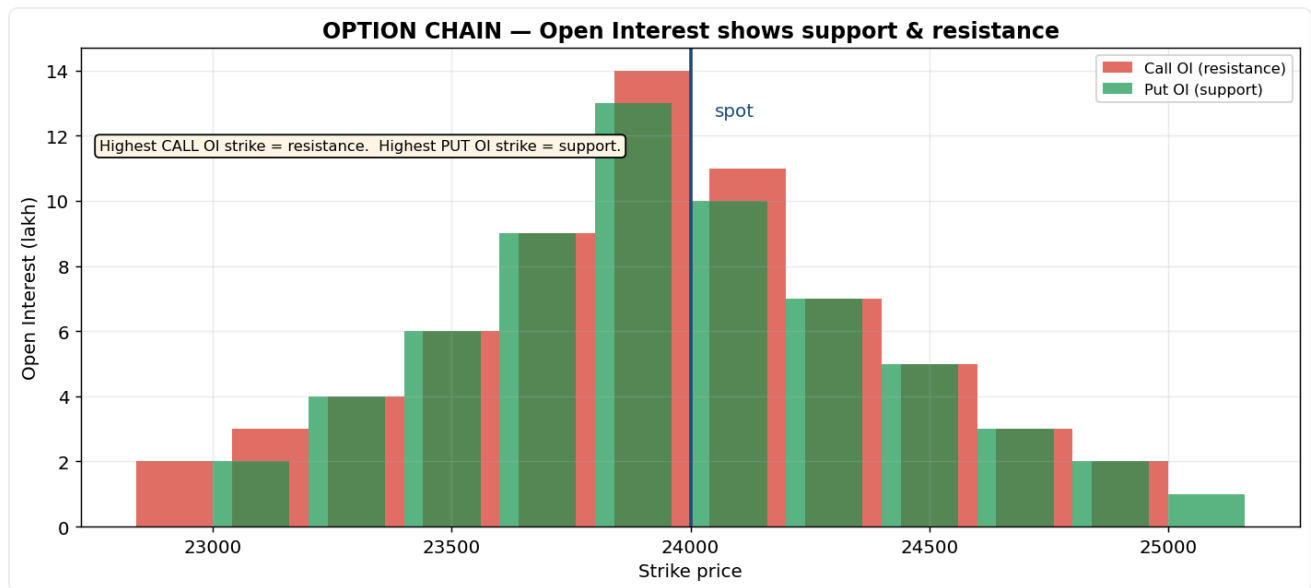
What it is: A bull call spread (buy one call, sell a higher one) caps cost and reward; an iron condor sells an OTM call spread and put spread.

Why it works: Spreads define risk precisely and lower cost; the condor monetises a market that goes nowhere.

The psychology: A spread trader accepts a capped reward to reduce cost and risk — a disciplined, probability-minded approach. The iron condor seller is betting on calm/range-bound markets and profits from time decay as long as price stays inside the range — selling 'insurance' to nervous traders.

How to use / trade it: Use spreads to express a view cheaply with known max loss; condors when you expect low volatility — but their risk is a sudden breakout.

Open Interest & the option chain



What it is: Open Interest = number of outstanding contracts at each strike, displayed across the option chain.

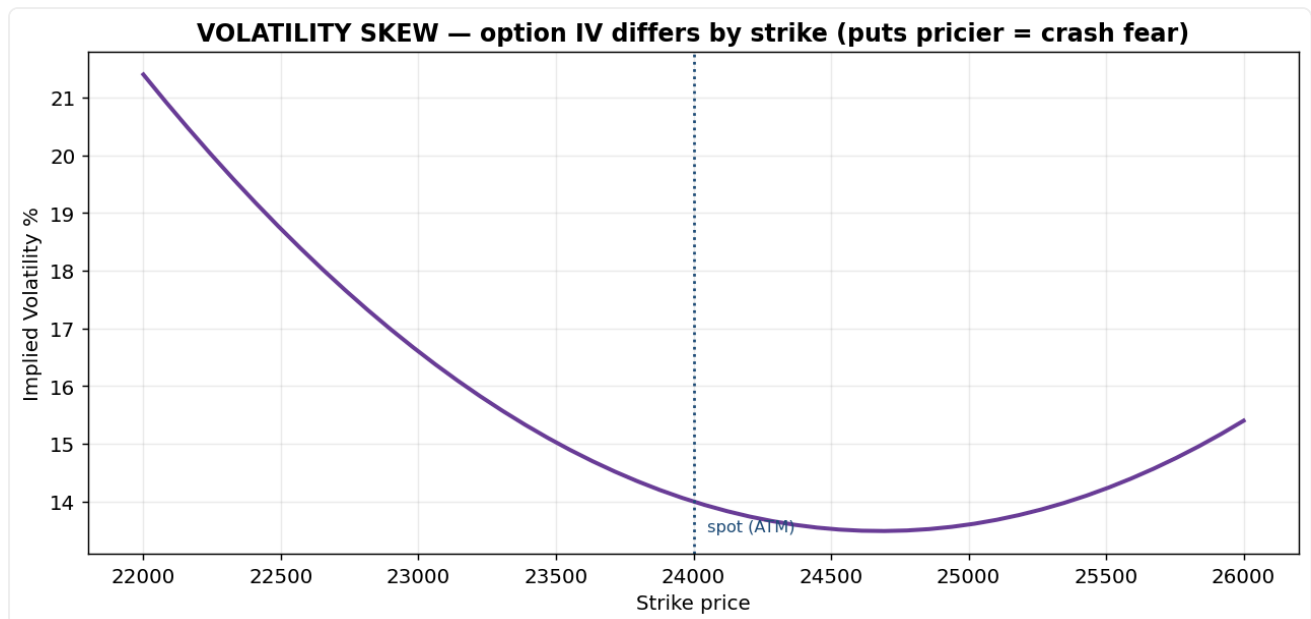
Why it works: It shows where large positions (often institutional writers) are concentrated, which creates magnetic support/resistance.

The psychology: The strike with the highest CALL open interest tends to act as resistance — call writers (who collected premium) defend it and want price to stay below. Highest PUT open interest acts as support for the same reason. **Max pain** is the strike where the most options expire worthless, and price often drifts there near expiry because large writers are incentivised to push it.

How to use / trade it: Read OI to map likely intraday/expiry support and resistance; rising OI + rising price = fresh longs (strong), rising OI + falling price = fresh shorts.

Interview depth: Mention PCR (put-call ratio) as a sentiment gauge — a very high PCR can be contrarian bullish (excessive fear).

Implied Volatility & skew



What it is: IV is the market's expected future volatility implied by option prices; skew = IV differing across strikes.

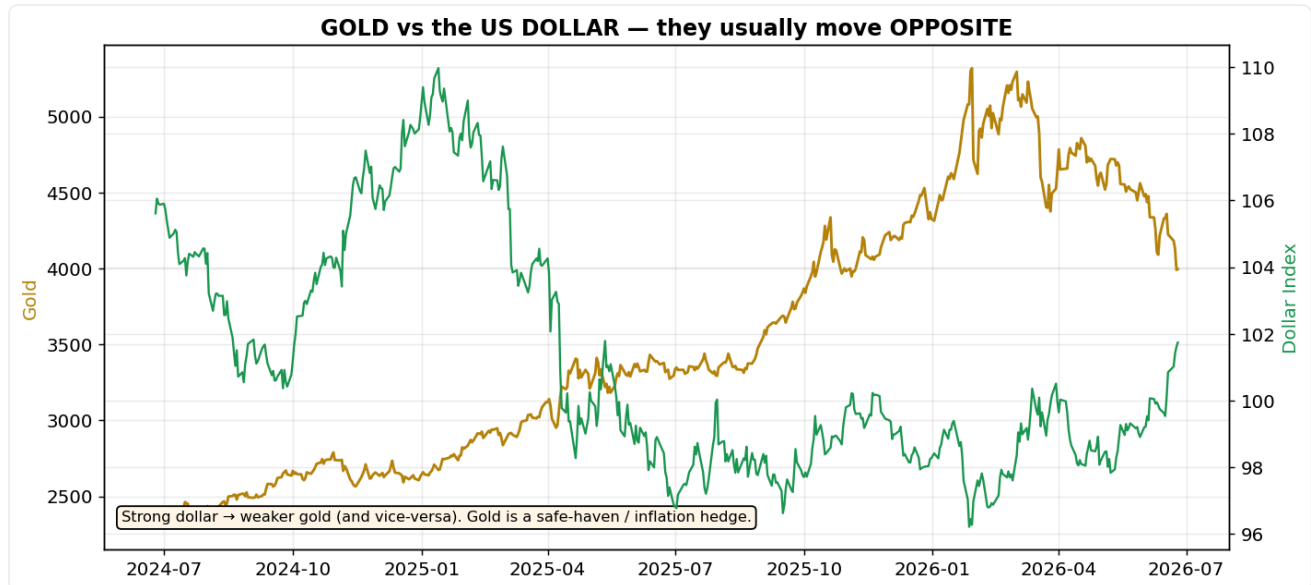
Why it works: Option prices reveal what the crowd expects volatility to be — fear and demand for protection get priced in.

The psychology: Downside puts usually carry higher IV (the 'skew') because investors fear crashes more than melt-ups and pay up for protection — a measurable fingerprint of fear. High overall IV means options are expensive (lots of expected movement / anxiety); low IV means complacency.

How to use / trade it: Buy options when IV is low and sell premium when IV is high; reading the IV surface (across strikes and expiries) is exactly the volatility-analysis work you did on the desk.

PART 6 & 7 — Commodities, Indices & Sentiment

Gold vs the US Dollar



What it is: Gold (priced in USD) and the US Dollar Index usually move in opposite directions.

Why it works: Because gold is priced in dollars and competes with the dollar as a store of value, a stronger dollar mechanically and psychologically weakens gold.

The psychology: Gold pays no interest, so when real yields/the dollar rise, money prefers cash/bonds and gold falls. In fear or high inflation, investors flee to gold as a safe haven and inflation hedge — gold rises on anxiety. It's a barometer of distrust in paper money.

How to use / trade it: Track the dollar, real yields and risk sentiment when forming a gold view; crude instead keys off OPEC supply, global demand and geopolitics.

India VIX — the fear gauge



What it is: VIX measures the market's expected 30-day volatility, derived from index option prices.

Why it works: When traders expect big moves they pay up for options, which pushes VIX up — so it reads collective anxiety.

The psychology: VIX spikes during selloffs because fear drives a rush to buy protective puts. Extreme VIX often coincides with capitulation bottoms (peak fear = sellers exhausted); very low VIX signals complacency and can precede sharp shocks.

How to use / trade it: Use VIX as a contrarian sentiment tool — spikes near support can flag bounce opportunities; persistently low VIX warns against complacency.

PART 8 — Risk Management (what makes a pro)

Risk : Reward and stops



What it is: Every trade defined by an entry, a stop-loss (exit if wrong) and a target, with reward sized larger than risk.

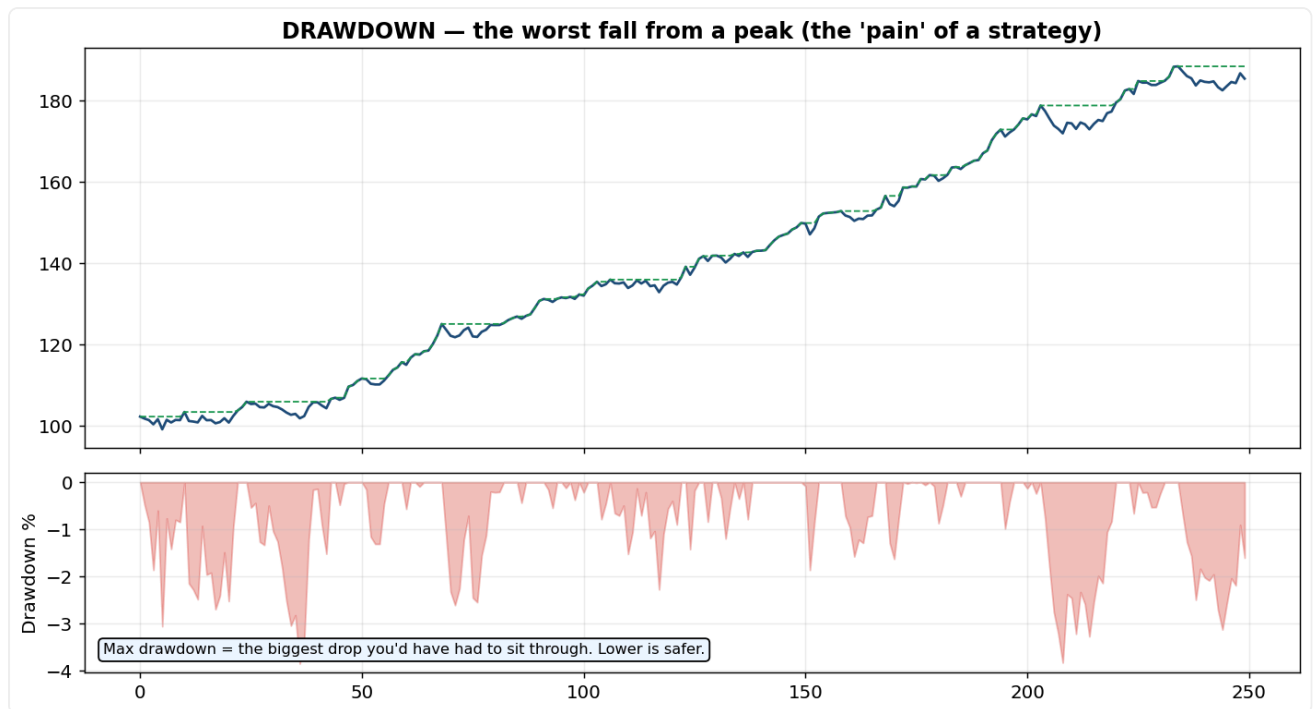
Why it works: Positive expectancy doesn't require being right often — it requires winners bigger than losers.

The psychology: The stop-loss exists to defeat loss aversion — the human urge to hold losers hoping they recover, which turns small losses into account-killers. Pre-deciding the exit removes emotion in the heat of the moment. A 1:2+ reward:risk means you can be wrong more than half the time and still grow capital.

How to use / trade it: Demand at least 2:1 reward:risk, set the stop at a level that invalidates your idea (not an arbitrary %), and never move a stop further away to avoid being wrong.

Interview depth: Frame it as 'survive first': controlling the downside is what keeps you in the game long enough for your edge to play out.

Drawdown — the pain measure



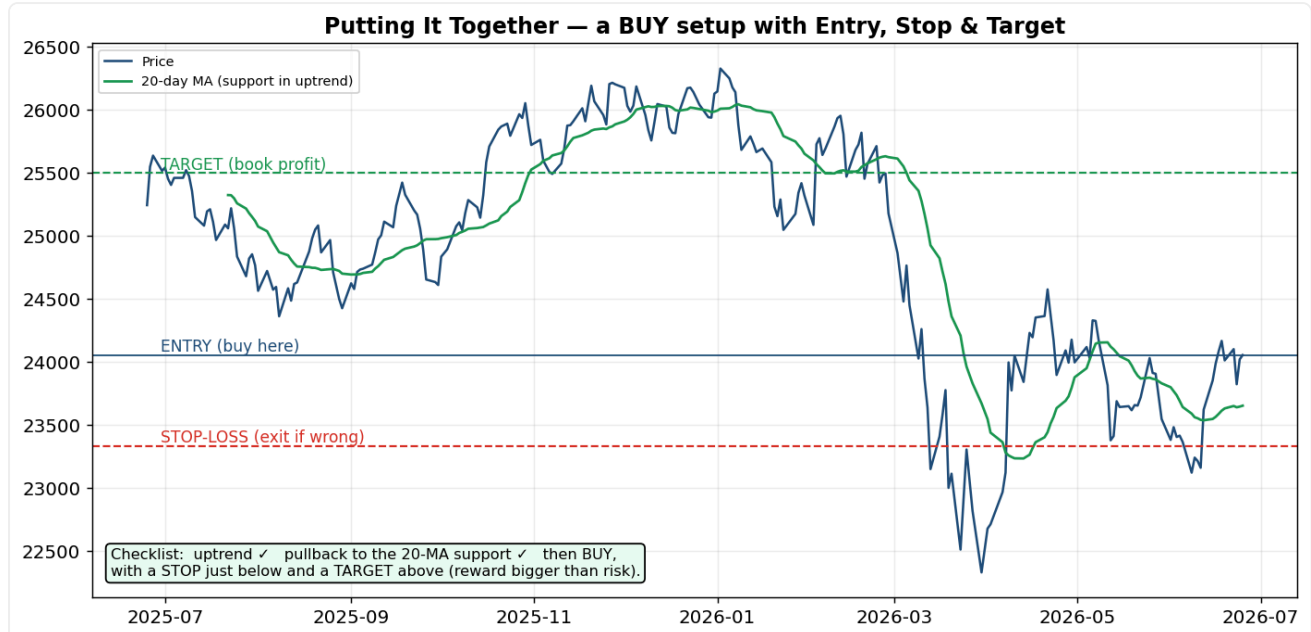
What it is: Drawdown is the fall from an equity peak; max drawdown is the worst such fall a strategy/account suffered.

Why it works: Returns alone are meaningless without knowing the pain endured to get them — drawdown measures risk of ruin and of quitting.

The psychology: Big drawdowns break traders psychologically — most abandon a strategy at its worst point, locking in the loss. A strategy with lower drawdown is one you can actually stick with, which is why risk-adjusted measures (Sharpe, Sortino) matter more than raw return.

How to use / trade it: Judge any strategy by drawdown and Sharpe/Sortino, not just total return; size positions so a normal drawdown is survivable both financially and emotionally.

Putting it ALL together — a real trade



What it is: A complete setup: confirm the uptrend, wait for a pullback to support/MA, then buy with a stop just below and a target above.

Why it works: It stacks independent edges — trend, level, momentum, and favourable risk:reward — so the probabilities line up in your favour.

The psychology: You're buying where the crowd that missed the move re-enters (support), in the direction the dominant players favour (trend), with a pre-planned exit that keeps emotion out. Each confluence factor is another group of traders acting in your favour.

How to use / trade it: Checklist every trade: trend ✓ → level ✓ → trigger (candle/indicator) ✓ → entry, stop, target with reward > risk. If any piece is missing, pass.

PART 9 — More Market Theory (Wyckoff, Sector Rotation, Breadth)

Wyckoff Method

Richard Wyckoff studied how "**smart money**" (**big institutions**) operate. His cycle: **Accumulation** → **Mark-up** → **Distribution** → **Mark-down** (the same four phases as the market cycle). The lesson: institutions accumulate quietly when the public is fearful and distribute quietly when the public is greedy. Watch for "**springs**" (a brief false breakdown below support that traps sellers right before a rally) — a classic Wyckoff accumulation signal.

Market cycles & sector rotation

The economy moves in cycles (early → mid → late → recession), and **different sectors lead at each stage**: financials & consumer discretionary early, industrials/energy/materials in the late boom, and defensives (FMCG, pharma, utilities) in downturns. **Relative strength** (comparing a stock or sector against the index) shows what is leading — analysts rotate coverage toward the leaders.

Market breadth

Breadth checks whether a move is broad or narrow: - **Advance/Decline line** — how many stocks are rising vs falling. - **% of stocks above their 200-day MA** — overall market health. - **New highs vs new lows**. A rally where the index rises but **breadth is weak** (few stocks participating) is fragile and often fails. Breadth confirms or warns against an index move.

PART 10 — Writing Research Reports & Making Calls

This is your actual daily output as a Technical Research Analyst.

Report cadence

- **Daily** — a pre-market note: overnight global cues, key levels for Nifty/Bank Nifty/commodities, and the top trade ideas.
- **Weekly** — trend review, sector view, positional calls.
- **Monthly** — the big-picture outlook and themes.

Anatomy of a good call

Every recommendation must state: **Instrument · View (Buy/Sell/Hold) · Entry · Target(s) · Stop-loss · Time horizon · Rationale**. The rationale ties together trend, levels, indicators, and (for F&O) OI/IV. **A call without a stop-loss is unprofessional** — always give one and an honest risk:reward.

Communicating to clients & RMs

- Be **clear and concise** — relationship managers relay your view to non-technical clients, so lead with the call, then the reason.
 - Talk in **probabilities, not certainties** ("favourable risk-reward to go long above X," never "this will go up").
 - **Track and own your calls** — record hits and misses; your credibility is your track record.
-

PART 11 — Regulation, NISM & Ethics

You're NISM-certified — know these basics cold.

NISM Series-XV: Research Analyst

The **SEBI-mandated certification** required to legally publish research or buy/sell recommendations in India. It covers technical analysis, fundamentals, valuation, and — importantly — **regulation and ethics**.

SEBI (Research Analysts) Regulations — key duties

- **Disclosure of conflicts** — reveal any holding or interest in a security you recommend.
- **No front-running** — never trade ahead of your own published research.
- **Separation** of research from the dealing/trading and investment-banking functions.
- **Rational basis & records** — every recommendation must be justified and documented.
- **Fair dealing** — no miselading claims, no guaranteed returns, suitability in mind.

Why it matters in interviews

Showing you understand **disclosures, conflict management, and that research must be compliant** signals maturity beyond just chart-reading.

PART 12 — Behavioural Finance & Market Psychology

Markets are driven by **fear and greed** — technical analysis is, at its heart, the study of crowd psychology.

The emotional cycle of a market

optimism → excitement → **euphoria (the top)** → anxiety → denial → fear → **panic / capitulation (the bottom)** → hope → optimism. Tops form at maximum greed; bottoms at maximum fear.

Common biases to know (and name)

- **Herd mentality** — following the crowd instead of the analysis.
- **FOMO** (fear of missing out) — chasing a move that's already run.
- **Loss aversion** — holding losers (hoping they recover) but cutting winners early; the single biggest destroyer of accounts.
- **Confirmation bias** — only seeing evidence that supports your existing view.
- **Anchoring** — fixating on a price ("I'll sell when it gets back to ₹X").
- **Recency bias** — over-weighting what just happened.
- **Overconfidence** — taking oversized risk after a few wins.

The professional edge

It isn't a secret indicator — it's **discipline**: a written plan, predefined stops, and the emotional control to follow them when the crowd panics. That is what separates a professional from a gambler.

PART 13 — Tools of the Trade & Common Setups

Tools

- **Charting:** TradingView (industry standard), broker terminals, MetaTrader, and pro terminals like Bloomberg / Refinitiv (Reuters).
- **Data & screening:** NSE/MCX option chains, screeners (e.g. Chartink), and Excel for tracking and models.
- **Coding (your edge):** Python (pandas, yfinance, matplotlib) for backtesting, screening and automating reports — exactly what your projects show.
- **Daily workflow:** pre-market prep → global cues + key levels → intraday monitoring → end-of-day report → tracking your calls.

Common trading setups (putting it all together)

- **Trend pullback** — in an uptrend, buy a dip to a moving average or support, with the trend.
- **Breakout** — enter when price clears a key level/range on rising volume; stop just inside the range.
- **Reversal** — trade a confirmed reversal pattern + divergence at a major level (higher risk).
- **Mean-reversion** — in a range, fade the extremes (buy oversold support, sell overbought resistance).
- **Momentum** — buy relative-strength leaders; ride until momentum fades.
- **Event/volatility (F&O)** — straddles/strangles around results or events when a big move is expected.

The analyst's mental checklist (use this on every idea)

1. What's the **trend** on the higher timeframe?
2. Where are the key **support/resistance** levels?
3. What do **price action / candles** say at those levels?
4. Do **indicators** (one trend + one momentum + volume) confirm? Any **divergence**?
5. For F&O: what do **OI, PCR and IV** say?
6. Define the trade: **entry, target, stop, risk:reward, horizon**.
7. **Size** the position to risk only a small fixed %.
8. Write the call **clearly with a rationale** — and **track** it.

PART 14 — The Story Behind Every Name (memory hooks)

Candlestick patterns (these are Japanese, ~300 years old — most names describe what the candle LOOKS like)

Name	Why it's called that — your memory hook
Doji	Japanese for "same time." The open and close happen at the same price → almost no body. Hook: <i>open = close = doji.</i>
Hammer	It looks like a hammer — small head (body) on top, long handle (lower wick). It "hammers out the bottom" of a fall. Hook: <i>a hammer bangs in the floor.</i>
Hanging Man	Same shape as a hammer but at a top — looks like a man hanging, body = head, long lower wick = dangling legs. Appears after a rise = danger.
Shooting Star	Small body at the bottom, long upper wick — looks like a star that shot up and is falling back down. Appears at tops. Hook: <i>a falling star = price about to fall.</i>
Marubozu	Japanese for "bald / shaven head" — a full body with no wicks (no hair/shadow) . Means total one-side control.
Engulfing	One big candle engulfs (swallows) the previous smaller one. Hook: <i>big fish eats small fish.</i>
Harami	Japanese for "pregnant." A small candle sits inside the prior big candle's body — looks like a pregnant belly (big = mother, small = baby).
Morning Star	The "morning star" is the planet Venus, which appears just before sunrise. It signals dawn = a new day = a bottom forming = bullish. Hook: <i>morning = sun rising = price rising.</i>
Evening Star	The "evening star" is Venus at dusk, before night falls. Signals darkness coming = a top = bearish. Hook: <i>evening = sun setting = price setting.</i>
Three White Soldiers	Three strong green candles marching upward like soldiers advancing = strong bullish.
Three Black Crows	Three red candles stepping down — crows = a bad omen , descending = bearish.
Tweezer Top/Bottom	Two candles with matching highs (or lows) — like the two equal tips of a pair of tweezers.

Big picture: candlesticks were invented by Japanese rice trader **Munehisa Homma** in the 1700s. The names are visual metaphors — so when you hear one, **picture the shape** and the meaning follows.

Chart patterns (also mostly "it looks like...")

Name	Memory hook
Head & Shoulders	The chart literally looks like a head with a shoulder on each side . Break the "neckline" = reversal.
Double Top (M) / Double Bottom (W)	The shape draws the letter M (two peaks = top) or W (two troughs = bottom).
Cup & Handle	Looks like a teacup (rounded bottom) with a small handle (dip) on the right, then breaks out.
Triangle	The converging trendlines form a triangle .
Flag / Pennant	A sharp move = the flagpole , then a small rectangle = the flag (or a tiny triangle = the pennant).
Wedge	Two lines tilting and converging like a wedge (doorstop) shape.

Indicators

Name	What the name literally means (so you remember what it does)
RSI — Relative Strength Index	Measures the strength of gains RELATIVE to losses . The name = the formula: average gain ÷ average loss.
MACD — Moving Average Convergence Divergence	It tracks how two moving averages converge (come together) and diverge (spread apart) . The name is what it computes.
Bollinger Bands	Named after the man who invented them — John Bollinger (1980s). "Bands" = the upper/lower lines. Hook: <i>a person's name</i> .
Fibonacci	Named after Leonardo Fibonacci , the medieval Italian mathematician whose number sequence (1,1,2,3,5,8,13...) produces the 61.8% "golden ratio."
Stochastic	From Greek <i>stokhos</i> = " <i>aim/guess</i> " — in statistics "stochastic" means probabilistic . It estimates where the close sits in its range.
ADX — Average Directional Index	An average of directional movement = an index of trend strength.
ATR — Average True Range	The average of the " True Range " (the full daily range including overnight gaps).
OBV — On-Balance Volume	Keeps a running balance of volume (adds on up days, subtracts on down days).
VWAP — Volume-Weighted Average Price	Exactly what it says: the average price weighted by volume .
Ichimoku	Japanese " <i>Ichimoku Kinko Hyo</i> " = " <i>one-glance equilibrium chart</i> " — see the whole balance at one glance . The shaded area = <i>Kumo</i> = "cloud."
Pivot Point	A " pivot " is the point something turns/balances on — the level price pivots around for the day.
Golden Cross / Death Cross	Golden = precious/good = bullish; Death = bad = bearish. The two MAs cross .
Parabolic SAR	SAR = " Stop And Reverse "; the dots trace a parabola (curve).

Theories named after people

Name	The person
Dow Theory	Charles Dow — co-founder of the Wall Street Journal & Dow Jones index.
Elliott Wave	Ralph Nelson Elliott — accountant who mapped market "waves" in the 1930s.
Wyckoff Method	Richard Wyckoff — early-1900s trader who studied "smart money."
Gann	W.D. Gann — trader famous for angles/time analysis.

Options & derivatives (the BEST stories for memory)

Name	Why it's named that
Call option	The right to "call" the asset to you — i.e., demand it / buy it. Hook: <i>call it IN = buy.</i>
Put option	The right to "put" the asset onto someone — i.e., sell/deliver it to them. Hook: <i>put it AWAY = sell.</i>
Straddle	You "straddle" both sides like sitting on a fence with one leg each side — buy a call AND a put at the same strike . Hook: <i>one leg each side = call + put.</i>
Strangle	Like a straddle but hands spread wider apart — the call and put are at different (OTM) strikes . Hook: <i>strangle = wider grip.</i>
Covered Call	The call you sold is "covered" because you own the stock to deliver. (If you don't own it, it's a risky "naked" call.)
Iron Condor	The profit zone is a wide flat plateau with two "wings" — like the broad wings of a condor (big bird). "Iron" = built from both calls and puts (a constructed, defined-risk frame).
Butterfly	The payoff diagram looks like a butterfly — a peak in the middle with a wing on each side.
Contango	Futures priced higher further out. Hook: <i>"Contango — Costs more, climbing up the curve."</i> (From an old London term for a fee to continue/defer settlement.)
Backwardation	Futures priced lower further out — the curve runs "backward." Hook: <i>backward = lower later.</i>
The Greeks (Delta, Gamma, Theta, Vega)	Greek letters for option sensitivities. Delta = change vs price. Theta = Time decay. Vega = Volatility (note: Vega isn't a real Greek letter — it was picked because V = Volatility).
VIX	Volatility Index . Hook: <i>V for Volatility = the fear gauge.</i>
Max Pain	The strike that causes the maximum pain (loss) to the most option buyers at expiry.
PCR — Put-Call Ratio	Simply puts ÷ calls (open interest) — a sentiment gauge.

PART 15 — Formulas & Worked Examples

You don't need to compute these by hand in the job (software does it). But knowing the formula lets you **explain what the number really means** — which is what an interviewer tests.

Moving Averages

SMA (Simple Moving Average) = just the plain average of the last n closes.

$$\text{SMA}(n) = (P_1 + P_2 + \dots + P_n) / n$$

Example: last 5 closes = 10, 12, 14, 16, 18 → $\text{SMA}(5) = 70 / 5 = \mathbf{14}$.

EMA (Exponential Moving Average) = weights recent prices more (reacts faster).

$$k = 2 / (n + 1) \quad (\text{the "smoothing factor"})$$
$$\text{EMA}_{\text{today}} = \text{Price}_{\text{today}} \times k + \text{EMA}_{\text{yesterday}} \times (1 - k)$$

Example: $n = 10 \rightarrow k = 2/11 = 0.182$. If yesterday's EMA = 100 and today's price = 110: $\text{EMA} = 110 \times 0.182 + 100 \times 0.818 = 20.0 + 81.8 = \mathbf{101.8}$.

RSI — Relative Strength Index (the one you asked for, step by step)

1. For each day, find the change vs the previous close.
2. Split into GAINS (up days) and LOSSES (down days, as positive numbers).
3. Average Gain = average of gains over 14 days
Average Loss = average of losses over 14 days
4. RS (Relative Strength) = Average Gain / Average Loss
5. $\text{RSI} = 100 - (100 / (1 + \text{RS}))$

Worked example: over 14 days, **Average Gain = 1.0** and **Average Loss = 0.5**.

$$\text{RS} = 1.0 / 0.5 = 2$$
$$\text{RSI} = 100 - (100 / (1 + 2)) = 100 - 33.3 = 66.7 \rightarrow \text{leaning overbought-ish}$$

Sanity checks (this is the intuition to say in an interview): - All up days, no losses → $\text{RS} = \infty \rightarrow \text{RSI} = \mathbf{100}$. - All down days, no gains → $\text{RS} = 0 \rightarrow \text{RSI} = \mathbf{0}$. - Gains = losses → $\text{RS} = 1 \rightarrow \text{RSI} = \mathbf{50}$ (neutral). So **above 70 = buyers strongly dominated recently (overbought); below 30 = sellers dominated (oversold)**.

MACD

$$\text{MACD line} = \text{EMA}(12) - \text{EMA}(26)$$
$$\text{Signal line} = \text{EMA}(9) \text{ of the MACD line}$$
$$\text{Histogram} = \text{MACD line} - \text{Signal line}$$

Example: $\text{EMA}_{12} = 1020$, $\text{EMA}_{26} = 1000 \rightarrow \text{MACD} = \mathbf{+20}$. If $\text{Signal} = 15 \rightarrow \text{Histogram} = \mathbf{+5}$ (positive = bullish momentum; MACD above signal).

Bollinger Bands

Middle band = SMA(20)
 σ (std dev) = standard deviation of the last 20 closes
Upper band = Middle + 2σ
Lower band = Middle - 2σ

Example: SMA20 = 100, $\sigma = 5 \rightarrow$ Upper = **110**, Lower = **90**. ($\pm 2\sigma$ statistically contains ~95% of recent prices, so touches are "stretched.")

ATR — Average True Range

True Range (TR) = the largest of these three:
(a) High - Low
(b) | High - Previous Close |
(c) | Low - Previous Close |
ATR = average of TR over 14 days

Example: High = 105, Low = 100, Previous Close = 102. TR = max(105-100=5 , |105-102|=3 , |100-102|=2) = **5**. (Using the prev close catches overnight **gaps**, which High-Low alone would miss.)

Stochastic Oscillator

%K = (Close - Lowest Low) / (Highest High - Lowest Low) $\times 100$ (over n, e.g. 14)
%D = 3-period average of %K

Example: Close = 50, lowest low = 40, highest high = 60 $\rightarrow$ %K = (50-40)/(60-40) $\times 100$ = **50**. (Near 100 = closing at the top of its range = strong.)

Pivot Points (intraday)

Pivot (P) = (High + Low + Close) / 3 $\leftarrow$ of the PREVIOUS day
R1 = 2P - Low S1 = 2P - High
R2 = P + (High - Low) S2 = P - (High - Low)

Example: yesterday H = 110, L = 100, C = 105 $\rightarrow$ P = 105. R1 = 2 $\times$ 105 - 100 = **110**, S1 = 2 $\times$ 105 - 110 = **100**, R2 = **115**, S2 = **95**.

Fibonacci Retracement

For an up-move from Low (L) to High (H): range = H - L
Retracement level (x%) = H - (x% $\times$ range)
Key levels: 23.6%, 38.2%, 50%, 61.8%

Example: L = 100, H = 200 $\rightarrow$ range = 100. 38.2% level = 200 - 38.2 = **161.8** ; 61.8% level = 200 - 61.8 = **138.2**. (The ratios come from the Fibonacci sequence: 8/13 $\approx$ 0.618, etc.)

VWAP & OBV

VWAP = $\Sigma(\text{Price} \times \text{Volume}) / \Sigma(\text{Volume})$ (cumulative through the day)

Example: 10 shares @100 and 20 shares @102 $\rightarrow$ (1000 + 2040) / 30 = **101.33**.

```
OBV:  if Close > prev Close → OBV = OBV + Volume
      if Close < prev Close → OBV = OBV - Volume
```

Backtesting / performance metrics (your backtester project)

```
Daily return r    = (Price_today - Price_yesterday) / Price_yesterday
CAGR              = (Ending Value / Starting Value)^(1 / years) - 1
Sharpe ratio      = (mean daily return / std dev of daily returns) × √252
Sortino ratio     = (mean daily return / std dev of NEGATIVE returns only) × √252
Max Drawdown     = max over time of (Peak - Trough) / Peak
Profit Factor     = gross profit / gross loss
Win rate          = winning trades / total trades
```

Worked examples: - **CAGR:** 100 → 200 over 4 years = $2^{(1/4)} - 1 = 1.189 - 1 = \mathbf{18.9\% \text{ per year}}$. - **Sharpe:** mean daily 0.0008, std 0.01 → $(0.0008/0.01) \times \sqrt{252} = 0.08 \times 15.87 = \mathbf{1.27}$ (good, >1). - **Max Drawdown:** equity goes 100 → 150 → 90. Peak = 150, trough = 90 → $(150-90)/150 = \mathbf{40\%}$. - **Profit Factor:** total winners +30, total losers -15 → $30/15 = \mathbf{2.0}$ (made ₹2 for every ₹1 lost). - (√252 annualises daily numbers because there are ~252 trading days a year.)

Options math

```
Intrinsic value (Call) = max(Spot - Strike, 0)
Intrinsic value (Put)  = max(Strike - Spot, 0)
Time value             = Premium - Intrinsic value
Long Call payoff@expiry = max(Spot - Strike, 0) - Premium
Long Put payoff@expiry  = max(Strike - Spot, 0) - Premium
Breakeven (long call)   = Strike + Premium
Breakeven (long put)    = Strike - Premium
PCR                     = Put Open Interest / Call Open Interest
```

Example: Buy a 24,000 call for ₹100 premium; at expiry Nifty = 24,250. Intrinsic = 24,250 - 24,000 = 250. Profit = 250 - 100 = **₹150 per unit**. Breakeven was 24,100. **Implied Volatility (IV):** there's no simple formula — it's **solved by reversing the Black-Scholes option-pricing model**: you know the option's market price and back out the volatility number that makes the model match. High IV = expensive options = market expects big moves.

How to use this sheet

1. **Part 1** — read each story once; the name will trigger the meaning in the interview ("Morning Star = Venus before sunrise = bottom = bullish").
2. **Part 2** — you don't memorise to compute by hand; memorise enough to **explain** ("RSI is average gain over average loss, scaled 0–100; 70+ means buyers dominated and it's overbought").
3. The killer combo: when asked about RSI, give the **formula + the sanity check** (all gains = 100, all losses = 0, equal = 50). That proves you understand it, not just recite it.

PART 16 — Interview Prep: Self-Intro, Rapid Revision & Q&A

0. Your 30-second self-intro (memorise the shape, not the words)

"I'm an MBA-Finance candidate and NISM Series-XV Research Analyst certified. I have close to two years on a derivatives desk at D.E. Shaw as Member Technical, where I analysed F&O market data — implied volatility, open interest, option greeks — and prepared daily reports for the trading team. Alongside that I do technical analysis on Nifty, Bank Nifty and commodities, and I've built tools to backtest strategies and generate daily research reports. I'm certified to publish buy/sell research and I want to do exactly that as a Technical Research Analyst."

PART A — What the job actually is

A **Technical Research Analyst (TRA)** studies **price charts** (not company balance sheets) to predict where Commodity / Futures & Options / equity prices are headed, and publishes **Buy / Sell / Hold calls** with a target and stop-loss in **daily, weekly and monthly reports**, and supports relationship managers (RMs) with market views.

Technical analysis vs Fundamental analysis (very common question): - **Fundamental** = *what to buy*. Studies company financials, earnings, valuation, economy. - **Technical** = *when to buy/sell*. Studies price & volume charts, assuming price already reflects all information and that trends and patterns tend to repeat.

The 3 assumptions of technical analysis (say these and you sound legit): 1. **Price discounts everything** — all news/info is already in the price. 2. **Prices move in trends** — once a trend starts it tends to continue. 3. **History repeats** — patterns recur because human behaviour (fear/greed) repeats.

PART B — The indicators ON YOUR RESUME (you MUST be able to explain each)

1. Trend

- **Uptrend** = higher highs + higher lows. **Downtrend** = lower highs + lower lows. **Sideways** = range.
- Golden rule: "*Trade with the trend.*" You buy dips in an uptrend, sell rallies in a downtrend.

2. Support & Resistance

- **Support** = a price *floor* where buyers repeatedly step in and stop the fall.
- **Resistance** = a price *ceiling* where sellers repeatedly step in and stop the rise.
- A **breakout** above resistance (or below support) signals a new move. Old resistance, once broken, often becomes new support (and vice-versa).
- *Interview line*: "I buy near support with a stop just below it, and target the next resistance."

3. Moving Averages (MA / DMA — Daily Moving Average)

- Average of the last N closing prices; smooths out noise to show the trend.
- **SMA** = simple average. **EMA** = exponential, weights recent prices more (reacts faster).
- Common ones: **20-DMA** (short term), **50-DMA** (medium), **200-DMA** (long term).
- Price **above** its MA = bullish; **below** = bearish.
- **Golden Cross** = 50-DMA crosses *above* 200-DMA → bullish. **Death Cross** = opposite → bearish.

4. RSI — Relative Strength Index

- A momentum oscillator from **0 to 100** measuring how strong recent gains are vs losses (default 14 days).
- **Above 70 = overbought** (may pull back). **Below 30 = oversold** (may bounce).
- **Divergence** = price makes a new high but RSI doesn't → momentum weakening, possible reversal.
- *Interview line*: "RSI tells me if a move is overstretched; I don't chase a stock at RSI 80."

5. MACD — Moving Average Convergence Divergence

- = (12-day EMA – 26-day EMA), plus a **9-day signal line**, plus a **histogram** of the gap.
- **Bullish** when the MACD line crosses *above* the signal line; **bearish** when it crosses below.

- Tells you momentum and trend direction together.

6. Bollinger Bands

- A 20-day moving average with an upper and lower band at **±2 standard deviations**.
- They measure **volatility**: bands widen when volatile, narrow ("squeeze") when calm.
- Price near the **upper band** = strong/overbought; near the **lower band** = weak/oversold.
- A **squeeze** (very narrow bands) often comes *before* a big breakout.

7. Volume

- Number of shares/contracts traded. **Volume confirms price**. A breakout on high volume is reliable; on low volume it's suspect.

8. Candlestick patterns (each candle shows Open, High, Low, Close)

- Green/white = close above open (up day). Red/black = close below open (down day).
- **Body** = open-to-close; **wicks/shadows** = the highs/lows.
- **Doji** = tiny body → indecision, possible reversal.
- **Hammer** = small body at top + long lower wick → bullish reversal after a fall.
- **Bullish Engulfing** = a big green candle fully covers the prior red candle → bullish reversal.
- **Shooting Star** = small body at bottom + long upper wick → bearish reversal.

9. Chart patterns (know the names + what they signal)

- **Head & Shoulders** = reversal (top = bearish). **Double Top / Bottom** = reversal.
- **Triangles / Flags / Pennants** = continuation (trend resumes after a pause).

10. Fibonacci retracement

- After a move, prices often pull back to **23.6%, 38.2%, 50%, 61.8%** of that move before continuing.
- These levels act as support/resistance. **61.8%** is the "golden ratio" and the most watched.

11. ATR — Average True Range

- Measures **how much an asset typically moves in a day** (volatility). Used to **size stop-losses**: a wider ATR → wider stop. *Interview line*: "I set my stop ~1.5×ATR away so normal noise doesn't hit it."

PART C — How to make a Buy/Sell call (the analyst's core job)

Every call has **4 parts**: 1. **View**: Buy / Sell / Hold. 2. **Entry**: the price to act at. 3. **Target**: where you book profit (often the next resistance, or a risk:reward multiple). 4. **Stop-loss**: where you admit you're wrong and exit (protects capital).

- **Risk:Reward** = $(\text{target} - \text{entry}) \div (\text{entry} - \text{stop})$. Aim for **at least 1:1.5 or 1:2**.
- *Example to say*: "Buy Bank Nifty at 58,400, target 60,300, stop 57,300 — that's roughly 1:1.7 reward-to-risk, and I'd exit if it closes below the 50-DMA."
- **Golden rule**: a good analyst is disciplined about **stop-losses**. "I'd rather take a small loss than a big one." Never average down a losing trade without a reason.

PART D — The markets you'll be asked about

Nifty 50 & Bank Nifty

- **Nifty 50** = index of the 50 largest NSE companies → the benchmark for the Indian market.
- **Bank Nifty** = index of the 12 largest banks → **more volatile**, moves faster, heavily traded in F&O.
- **Sensex** = BSE's 30-stock equivalent of Nifty.

Futures & Options (F&O / Derivatives) — you worked on this desk

- A **derivative** derives its value from an underlying (stock/index/commodity).
- **Futures** = an *obligation* to buy/sell at a set price on a future date.
- **Options** = a *right* (not obligation):

- **Call option** = right to **buy** (you buy calls if you're bullish).
- **Put option** = right to **sell** (you buy puts if you're bearish).
- **Strike price** = the agreed price. **Premium** = what you pay for the option. **Expiry** = last day.
- **Open Interest (OI)** = total open contracts; rising OI + rising price = strong trend.
- **Implied Volatility (IV)** = the market's expectation of future volatility, priced into options. High IV = expensive options. (You analysed IV surfaces — be ready to say "IV across strikes/expiries".)
- **Option Greeks** (you tracked these):
 - **Delta** = how much the option moves per ₹1 move in the underlying.
 - **Gamma** = how fast delta changes.
 - **Theta** = time decay (options lose value as expiry nears).
 - **Vega** = sensitivity to volatility.
- **Strategies: Straddle** (buy call+put same strike — bet on a big move either way), **Strangle** (same idea, different strikes, cheaper), **Spread** (buy+sell options to cap risk/cost).

Commodities (MCX) — the JD's first bullet

- Traded on **MCX** (Multi Commodity Exchange) in India: **Gold, Silver, Crude Oil**, natural gas, copper.
- Indian prices track **global benchmarks**: Gold → COMEX (USD/ounce), Crude → WTI/Brent (USD/barrel).
- **Drivers**: the **US dollar** (stronger USD → weaker gold), **interest rates & inflation** (gold is an inflation hedge / safe haven), **geopolitics & supply** (crude reacts to OPEC, wars), and **demand**.
- Same technical tools (S/R, RSI, MACD, MAs) apply to commodity charts.

PART E — NISM & the rules (you're certified — know the basics)

- **NISM Series-XV: Research Analyst** is the SEBI-mandated certification to legally publish research / buy-sell recommendations in India.
- Key principles a Research Analyst must follow:
 - **Disclosure** of any holding/conflict of interest in a stock you recommend.
 - **No front-running** — you can't trade ahead of your own published call.
 - **Separation** of research from the dealing/trading function.
 - Recommendations must have a **rational basis** and be documented.
- *Interview line*: "Being NISM-certified means I can publish compliant research with proper disclosures and a documented rationale."

PART F — How to explain YOUR 3 projects (they WILL ask)

1. Daily/Weekly/Monthly Technical Research Reports (Nifty, Bank Nifty, Gold, Crude):

"I prepare a morning note on indices and commodities — I mark support/resistance, check RSI, MACD, Bollinger Bands and candlestick patterns, and give a Buy/Sell/Hold call with entry, target and an ATR-based stop-loss. I automated the data part in Python so the levels stay current."

2. Multi-Strategy Technical Backtester (Python):

"I backtested four technical strategies — MA crossover, RSI mean-reversion, MACD trend and Bollinger breakout — on six years of Nifty and Bank Nifty data, and compared them to buy-and-hold on CAGR, Sharpe, Sortino, max drawdown, win-rate and profit factor. The active strategies underperformed buy-and-hold in a strong bull market, but **MACD-trend gave the best risk-adjusted return with the smallest drawdown** — the real lesson was validating strategies with data instead of trusting them blindly." *(This honesty + the metrics vocabulary impresses interviewers.)*

Backtesting metrics — be able to define these (they're in your report): - **CAGR** = the smoothed annual growth rate of the strategy (return per year). - **Sharpe ratio** = return per unit of *total* risk (volatility). Higher is better; >1 is good. - **Sortino ratio** = like Sharpe but only penalises *downside* volatility — a fairer risk measure. - **Max Drawdown** = the worst peak-to-trough fall the strategy suffered (how much pain you'd endure). - **Win-rate** = % of trades that were profitable. - **Profit Factor** =

gross profit ÷ gross loss; above 1 means the system made money overall. - **Exposure** = % of the time the strategy was actually invested (rest in cash).

3. Multi-Asset Technical Coverage Watchlist:

"A ranked watchlist of 10 NSE stocks plus Gold and Crude, scored on trend, momentum and candlestick signals, tagging Buy/Sell/Hold with stop-losses — the kind of coverage sheet you'd hand to RMs each morning."

PART G — 20 likely questions (one-line answers to expand on)

1. **What is technical analysis?** → Studying price/volume charts to forecast prices; price discounts everything.
2. **Technical vs fundamental?** → Fundamental = what to buy; technical = when to buy/sell.
3. **What is support & resistance?** → Floor where buyers step in / ceiling where sellers step in.
4. **Explain RSI.** → 0–100 momentum; >70 overbought, <30 oversold.
5. **Explain MACD.** → 12-EMA minus 26-EMA with a 9-signal line; cross above = bullish.
6. **What are moving averages? Golden cross?** → Smoothed avg price; 50 crossing above 200 = bullish.
7. **Bollinger Bands?** → 20-MA ±2 SD; measures volatility; squeeze precedes breakout.
8. **What's a candlestick? Name a reversal pattern.** → OHLC bar; hammer / bullish engulfing.
9. **Fibonacci retracement levels?** → 23.6/38.2/50/61.8%; pullback support/resistance.
10. **How do you decide a buy?** → Trend up + price above MAs + RSI not overbought + MACD bullish + near support.
11. **What is a stop-loss and why?** → A pre-set exit to cap losses; protects capital and emotions.
12. **Risk-reward ratio?** → Reward ÷ risk; want ≥1:2.
13. **Difference between Nifty and Bank Nifty?** → 50-stock benchmark vs 12 banks, Bank Nifty more volatile.
14. **Call vs Put option?** → Right to buy vs right to sell.
15. **What is Open Interest / Implied Volatility?** → Open contracts / market's expected future volatility.
16. **Name the option greeks.** → Delta, Gamma, Theta, Vega.
17. **What's a straddle?** → Buy call + put at same strike to profit from a big move either way.
18. **What drives gold / crude prices?** → USD, rates, inflation, safe-haven / OPEC, supply-demand, geopolitics.
19. **What does NISM RA certification allow?** → To publish compliant research with disclosures.
20. **Walk me through one of your buy calls.** → Use the Bank Nifty example in Part C.

PART H — Smart questions to ASK them (shows interest)

- "Which segments will I cover — equity, F&O, or commodities — and on what timeframes?"
- "Do analysts here publish their own calls or support senior analysts first?"
- "What tools/terminals does the desk use — TradingView, broker terminal, Bloomberg?"
- "How is a call's performance tracked here?"

Dos & Don'ts

- ☒ Speak in terms of **trend, levels, risk-reward, stop-loss** — that's the analyst's language.
- ☒ Admit what you don't know briefly, then pivot to what you do.
- ☒ Always attach a **stop-loss** to any call you mention — never sound reckless.
- ☒ Don't claim you made live buy/sell calls at D.E. Shaw — you were **Member Technical** (analysis/support).
- ☒ Don't guarantee returns or say "this will definitely go up." Analysts talk in probabilities and risk.

Tonight's priority order (if short on time)

1. Part B (indicators) + Part C (buy/sell call) — **non-negotiable**, it's your core.
2. Part A (TA vs fundamental, 3 assumptions) — easy marks.
3. Part D (Nifty/Bank Nifty, options basics, commodity drivers).
4. Part F (your projects) + Part G (Q&A) — rehearse out loud.
5. Part E (NISM) + Part H — quick skim.

Good luck. You know more than you think once you read this twice.